

|                                                                                                                                                                                                                                     |                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
|  <b>KIMS ALSHIFA</b><br><small>SUPER SPECIALITY HOSPITAL</small>  | Doc No: KASPP/AB/01                      |
|                                                                                                                                                                                                                                     | Date of Creation:20/05/2024              |
|                                                                                                                                                                                                                                     | Date of Approval:30/06/2024              |
|                                                                                                                                                                                                                                     | Date of Issue:30/06/2024                 |
|                                                                                                                                                                                                                                     | Revision Date and Status: 01/08/2024 /04 |
| <b>Hospital Infection Control</b>                                                                                                                                                                                                   | Date of Review:01/08/2025                |
| <b>ANTIBIOTIC POLICY</b>                                                                                                                                                                                                            | Page 1 of 98                             |

# ANTIBIOTIC POLICY

This Document is part of the Departmental Manual of Infection Prevention & Control,  
KIMS Alshifa Hospital

## Hospital Infection Control

### ANTIBIOTIC POLICY

## PREFACE

Antibiotics are powerful medications that have revolutionised the treatment of infectious diseases and saved countless lives. However, the misuse and overuse of antibiotics have led to the emergence of antibiotic resistance, which poses a significant threat to public health. Antibiotic resistance is a global health crisis where bacteria become resistant to the drugs designed to kill them. Misuse and overuse of antibiotics contribute to the development of antibiotic-resistant bacteria. But in recent scenarios, Antimicrobial Resistance (AMR) is a major global health concern that threatens the effective prevention and treatment of a wide range of infectious diseases, as antimicrobial drugs become less effective; infections that were once easily treatable may become difficult or impossible to control. This can lead to longer illness durations, increased complications, and higher mortality rates.

A state-level antibiotic guideline may be an effective document to guide healthcare professionals in the appropriate and responsible use of antibiotics, ensuring their effectiveness while minimising the development of resistance. An antibiotic policy provides guidelines and protocols for healthcare professionals regarding the appropriate use of antibiotics. It emphasises the importance of prescribing antibiotics only when necessary, selecting the right drug and dose, and ensuring the proper duration of treatment. By promoting responsible antibiotic use, the policy helps to minimise the selective pressure on bacteria, reducing the likelihood of resistance development. It also helps combat Antimicrobial Resistance by discouraging inappropriate prescribing practices.

All recommended therapies are either evidence-based as per universally accepted standards or based on available antibiograms at different levels of hospitals, including our own. These are general guidelines; treatment of individual patients may vary depending on local conditions and experience. The antimicrobial susceptibility data given in this guideline are available through the online portal KONTENT.

Antimicrobial resistance data is known to differ between different healthcare institutes and even between different wards like indoor, critical care units, etc., of the same institute. These guidelines will help clinicians, to implement antibiotics, mainly empirical antibiotic therapy, before receiving the culture and sensitivity reports.

**Hospital Infection Control**
**ANTIBIOTIC POLICY**
**CONTENTS**

| <b>SI NO</b> | <b>TOPIC</b>                                                | <b>PAGE NO</b> |
|--------------|-------------------------------------------------------------|----------------|
| <b>1</b>     | <b>Antimicrobial Stewardship Programme</b>                  | <b>3</b>       |
| <b>1.1</b>   | <b>Definition</b>                                           | <b>3</b>       |
| <b>1.2</b>   | <b>Objectives</b>                                           | <b>3</b>       |
| <b>1.3</b>   | <b>Effective intervention for Antimicrobial Stewardship</b> | <b>3</b>       |
| <b>1.4</b>   | <b>Role of IPCC &amp; ICT in Antimicrobial Stewardship</b>  | <b>4</b>       |
| <b>1.5</b>   | <b>Antimicrobial Stewardship Program</b>                    | <b>5</b>       |
| <b>1.6</b>   | <b>General Antibiotic Use Guidelines</b>                    | <b>6</b>       |
| <b>1.7</b>   | <b>Steps to Follow the Antibiotic Protocol</b>              | <b>9</b>       |
| <b>1.8</b>   | <b>Patient Risk Stratification</b>                          | <b>10</b>      |
| <b>1.9</b>   | <b>Antibiotic Protocol – OPD &amp; IP</b>                   | <b>11</b>      |
| <b>1.10</b>  | <b>Antibiotic Protocol – ICU</b>                            | <b>12</b>      |
| <b>1.11</b>  | <b>Categorization of Antibiotics</b>                        | <b>12</b>      |
| <b>1.12</b>  | <b>Restricted Use Antibiotics</b>                           | <b>13</b>      |
| <b>1.13</b>  | <b>Limited Access Antibiotics</b>                           | <b>14</b>      |
| <b>1.14</b>  | <b>Under Surveillance Antibiotics</b>                       | <b>15</b>      |
| <b>1.15</b>  | <b>Irrational combinations/less evidenced combinations</b>  | <b>15</b>      |
| <b>1.16</b>  | <b>Peri-Surgical Antibiotic Use</b>                         | <b>16</b>      |
| <b>1.17</b>  | <b>Surgical Prophylaxis Guidelines</b>                      | <b>19</b>      |
| <b>1.18</b>  | <b>Wound Classification Algorithm</b>                       | <b>29</b>      |
| <b>2</b>     | <b>Empiric Treatment of Common Infection</b>                | <b>30</b>      |
| <b>2.1</b>   | <b>Gastro-Intestinal Infections</b>                         | <b>30</b>      |
| <b>2.2</b>   | <b>Genito-Urinary Infections</b>                            | <b>37</b>      |
| <b>2.3</b>   | <b>Neurological Infections</b>                              | <b>43</b>      |

## Hospital Infection Control

### ANTIBIOTIC POLICY

|      |                                                |     |
|------|------------------------------------------------|-----|
| 2.4  | Orthopedic Infections                          | 47  |
| 2.5  | Respiratory Infections                         | 50  |
| 2.6  | Sepsis & Other Blood Stream Infection          | 73  |
| 2.7  | Skin & Soft Tissue Infection                   | 76  |
| 2.8  | Other Infections                               | 87  |
| 2.9  | Obstetrics & Gynecological Infection           | 91  |
| 2.10 | Pediatric Infections Dosages                   | 100 |
| 2.11 | Pediatric Infections                           | 102 |
| 2.12 | Antimicrobial Dosing in Renal Insufficiency    | 119 |
| 3.1  | Annexure 1: Pre-authorisation form             |     |
| 3.2  | Annexure 2: Rational Antibiotic use monitoring |     |

## 1. ANTIMICROBIAL STEWARDSHIP PROGRAMME

A multifaceted approach is necessary to prevent, detect, and control the emergence of antimicrobial-resistant organisms. This includes ensuring the availability of adequate and appropriate therapeutic agents, the existence of diagnostic capacity to rapidly and reliably detect specific pathogens and their antimicrobial susceptibilities and the promotion of robust infection prevention, control, and antimicrobial stewardship programs. This policy focuses on issues relating to antimicrobial stewardship.

### 1.1 Definition

Antimicrobial stewardship refers to coordinated interventions designed to improve and measure the appropriate use of antimicrobial agents by promoting the selection of the optimal antimicrobial drug regimen including dosing, duration of therapy and route of administration.

## Hospital Infection Control

### ANTIBIOTIC POLICY

## 1.2 Objectives

The major objectives of antimicrobial stewardship are to achieve best clinical outcomes related to antimicrobial use while minimizing toxicity and other adverse events, thereby limiting the selective pressure on bacterial populations that drives the emergence of antimicrobial-resistant strains. Antimicrobial stewardship may also reduce excessive costs attributable to suboptimal antimicrobial use.

## 1.3 Effective interventions for Antibiotic Stewardship

**1.3.1** When establishing a new stewardship program, it is best to start with the core strategies and focus on achieving and maintaining them before adding some of the supplemental strategies.

### 1.3.1.1 Core Strategies

- I. Front-end strategies where antimicrobials are made available through an approval process (e.g. formulary restrictions and preauthorization)
- II. Back-end strategies are where antimicrobials are reviewed after antimicrobial therapy has been initiated (e.g. prospective audit with intervention and feedback).

### 1.3.1.2 Supplemental strategies

- I. Streamlining / timely de-escalation of therapy
- II. Dose optimization
- III. Parenteral to oral conversion
- IV. Guidelines and clinical pathways
- V. Antimicrobial order forms

## Hospital Infection Control

### ANTIBIOTIC POLICY

- VI. Education
- VII. Computerized decision support,
- VIII. Surveillance
- IX. Laboratory surveillance and feedback
- X. Combination therapies
- XI. Antimicrobial cycling

## 1.4 Role of IPCC & ICT in Antibiotic Stewardship

- 1.4.1** The IPCC will assist the Pharmacy and Therapeutics Committee (PTC) in finalizing the list of antibiotics in the hospital formulary.
- 1.4.2** The ICT along with Clinical Pharmacist will act as the antibiotic management team for daily monitoring of antibiotic use.
- 1.4.3** Antibiotic policy shall be prepared by the Infection Control Physician (ICP) in consultation with IPCC. The policy shall be reviewed and updated annually.

## 1.5 Antimicrobial Stewardship Program

### 1.5.1 Antimicrobial Stewardship Program monitoring activities shall include:

- I. Rational use of antibiotics shall be monitored on daily basis for restricted use indicator antibiotics (*Vancomycin*, *Meropenem*, *Colistin*, *Levofloxacin*, *Daptomycin*, *Tigecycline*, *Ceftaroline* and non-TB use of rifampicin or any other antibiotic outside hospital formulary) by ICT on daily rounds and details recorded on preformatted template. Other antibiotics are also checked for rational combinations, doses and duration prescribed. Treating doctors shall be asked to explain the reasons for

## Hospital Infection Control

### ANTIBIOTIC POLICY

initiating these antibiotics in writing or ICP shall discuss the rationality with them directly. Irrational antibiotic therapy, if identified is communicated to treating physician or surgeon for immediate discontinuation/modification. Irrational combination of antibiotics or doses is also monitored. The continued need for antimicrobial therapy should be reviewed at least daily. For most types of infection treatment should continue until the clinical signs and symptoms of infection have resolved. Parenteral therapy is normally used in seriously ill patients and those with gastrointestinal upset. Oral therapy can often be substituted as the patient improves.

- II. Pre-surgical prophylaxis and post-operative antibiotic therapy shall be monitored on daily basis whether in line with peri-surgical antibiotic prophylaxis guidelines. In case of irrationality concerned department is informed and necessary actions are taken.
- III. Defined Daily Dose (DDD) or Days of Therapy (DOT) for antibiotics per thousand days are calculated and monitored for the antibiotic usage pattern by the ICT in coordination with Clinical Pharmacist.
- IV. Number of doses administered are also monitored per thousand patient days.
- V. The data analysis is done and discussed periodically during IPCC meetings and feedback provided to the users.
- VI. Adherence to antibiotic policy shall also be discussed in the IPCC meeting.
- VII. Prescription audits of inpatients and outpatients shall be

## Hospital Infection Control

### ANTIBIOTIC POLICY

conducted periodically by the Clinical Pharmacist in co-ordination with ICT.

- VIII. ABC analysis of medicines shall be done annually by drug stores and records are to be submitted to IPCC at least once in six months.
- IX. Antibiotic usage shall be monitored particularly if any of the seven indicator antibiotics come in the top 10 medicines.

### 1.5.2 Aims of Antimicrobial Stewardship

- I. To provide a simple, best empirical/specific treatment of common infections
- II. To promote the safe, effective, economic and rational use of antibiotics
- III. To minimize the emergence of bacterial resistance in the community

## 1.6 General antibiotic use guidelines

- I. All antibiotic initiations shall be done after sending appropriate samples for cultures or any changes in antibiotic is done after receiving culture report.
- II. Rapid tests e.g. Gram stain shall be done to determine therapeutic choices when decision on empiric therapy is required.
- III. The IPCC shall categorize usage of antibiotics for restricted use, limited access and under surveillance based on antibiogram, if available and/or in consultation with IPCC of the Hospital.
- IV. KIMS Al Shifa hospital shall have a list of antibiotics made available for OPD, IP, emergency and respective ICUs in consultation with IPCC of the Hospital.



## Hospital Infection Control

### ANTIBIOTIC POLICY

- V. List of all available antibiotics shall be communicated to the prescribers every month or from time-to-time if there is any change in the list or medicine is not available for some reasons by the Clinical Pharmacist.
- VI. Antimicrobials shall be chosen following hospital policy and national standard treatment guidelines for infectious diseases. If alternatives are chosen, reason for the same shall be documented in the case records.
- VII. Antibiotic shall be prescribed only when there is likely to be a clear clinical benefit.
- VIII. Antibiotic shall not be prescribed for viral sore throat, simple coughs and colds and viral diarrhea.
- IX. Empiric therapy shall be given where delay in initiating therapy awaiting microbiological results would be life threatening or risk serious morbidity, antimicrobial therapy based on a clinically defined infection is justified. Necessary specimens are drawn before commencing therapy. Where empiric therapy is used the accuracy of diagnosis shall be reviewed regularly and treatment altered/stopped when microbiological results become available.
- X. Once culture & sensitivity report are available, presumptive therapy antibiotic may require to be changed. Consult Microbiologist to decide the choice of antibiotic (based on narrowest spectrum antibiotic which covers the pathogen isolated).
- XI. Following factors affecting antimicrobial choices and route of administration shall be checked by the prescriber and ICT e.g., age, type and site of infection (respiratory, intra-abdominal, pneumonia, blood stream,

## Hospital Infection Control

### ANTIBIOTIC POLICY

urinary tract and skin and soft tissue), renal & hepatic function, interactions, allergy, if any.

- XII. A dose and duration of treatment shall be suggested by the ICP but can be modified by consultants based on clinical scenarios.
- XIII. Use simple generic antibiotics first whenever possible. Avoid broad spectrum antibiotics (e.g. Amoxycillin+Clavulanate, Quinolones and Cephalosporins) when standard and less expensive antibiotics remain effective, as they increase the risk of Clostridium difficile, MRSA and resistant UTIs.
- XIV. Avoid widespread use of topical antibiotics.
- XV. All allergies shall be recorded prominently in red ink in the allergy box on the patient's case sheet. Drug chart is completed when a new prescription chart is written or transcribed. If no allergy - "No known allergy or allergic to name of the drug" is recorded. The box is signed and dated. If allergy history cannot be obtained, then "history not available" is specified. Under no circumstances allergy box shall be left blank. The allergy box is completed before prescribing a new drug, except in exceptional circumstances. If patients have a suspected drug allergy then the drug and suspected reaction shall be documented in the case sheet and the drug chart.
- XVI. Check that the appropriate dose is prescribed. If uncertain, contact ICP, Pharmacy or check in the formulary.
- XVII. The need for antimicrobial therapy shall be reviewed daily. For most types of infection, treatment is continued until the clinical signs and symptoms of infection have resolved – exceptions to this shall be considered if

## Hospital Infection Control

### ANTIBIOTIC POLICY

documented in standard guidelines or published article.

For most infections 5-7 days of

antimicrobial therapy is sufficient (simple UTIs can be adequately treated with 3 days of antibiotic).

XVIII. Parenteral therapy is normally used in seriously ill patients and those with gastrointestinal upset. All IV antibiotics are initially given for 48 – 72 hours without review and switching over to oral alternatives is considered after 48 hours. Oral therapy can often be substituted as the patient improves. Switching to oral is indicated by fever defervescence for at least 24h, marked clinical improvement and low CRP or PCT.

XIX. Antimicrobials are de-escalated or stepped down to the narrowest spectrum, most efficacious and most cost-effective option as per culture reports. If no step down is available, the reason is documented and is subjected to clinical audit.

XX. Where treatment is apparently failing, advice from the Microbiologist is sought rather than blindly changing to an alternative choice of antimicrobial agent.

## 1.7 Steps to follow the Antibiotic Protocol

- I. Identify the type of infection — bloodstream, respiratory, intra-abdominal or urinary tract
- II. Define the location — OPD, ICU or IP patient
- III. Identify the patient type based on described parameters — Type 1, Type 2 or Type 3. Refer to the empiric/specific therapy for that patient type 1, 2 or 3 with first, second or third line antibiotic respectively.
- IV. Wait for at least 48 h of antimicrobial therapy before

## Hospital Infection Control

### ANTIBIOTIC POLICY

labelling patient as non- responding to the therapy and to switch to the higher next line of therapy. Also consider if patient condition deteriorates.

- V. Send respective cultures and/or primary set of investigations before starting antibiotic therapy
- VI. Once Culture & Sensitivity report are available initiate specific antimicrobial therapy. Antimicrobial may require to be changed/de-escalated.

## 1.8 Patients risk stratification

| PATIENT TYPE 1 (CAI)                         | PATIENT TYPE 2 (HAI)                                                                                                | PATIENT TYPE 3 (NI)                                                                                 |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| No contact with health care system           | Contact with health care system (e.g. Recent hospital admission, nursing home, dialysis) without INVASIVE procedure | Long hospitalization and or invasive procedure                                                      |
| No prior antibiotic treatment                | Recent antibiotic therapy                                                                                           | Recent & multiple antibiotic therapies                                                              |
| Patient young with few co- morbid conditions | Patient old with multiple co- morbidities                                                                           | Cystic fibrosis, structural lung disease, advanced AIDS, neutropenia, other severe immunodeficiency |

**\*Patient Type 4** - Patients with suspected INVASIVE fungal infection.

## 1.9 Antibiotic Protocol - OPD & IP

- I. For treating the IP, the microbiology data should be

## Hospital Infection Control

### ANTIBIOTIC POLICY

considered mainly for patients belonging to Patient Types 2 and 3. Patient type 1 refers to the patients reporting with community acquired infections, and for them the treatment options are based on the guidelines.

- II. Avoid Antipseudomonal Fluoroquinolones (e.g., ciprofloxacin and levofloxacin) and Antipseudomonal 3rd generation Cephalosporins (e.g. ceftazidime and cefoperazone) in patients Type 1 and 2 since they have potent antipseudomonal activity.
- III. Treatment of serious infections caused by ESBL+ve pathogens with BL-BLI combinations (e.g., Piperacillin-Tazobactam, Cefepime-Tazobactam and Cefoperazone-Sulbactam) should be avoided as there is limited clinical data and evidence available for the use of these drugs in such infections.
- IV. OPAT - Out Patient Parenteral Antibiotic Therapy allows patients requiring intravenous antibiotics to be treated outside hospital but is otherwise stable and well enough not to be in hospital. These patients may be discharged early to an OPAT service or may avoid hospital admission altogether. Early OPAT programs facilitate the discharge of stable in-patients with infections who, other than the requirement for prolonged intravenous antibiotic therapy, had no other need for inpatient care. OPAT are suitable for many infections, especially cellulitis, bone and joint infections, and infective endocarditis. Antibiotics can be administered in an outpatient unit, at home by a nurse or at home by the patient or a caregiver, however, patients should be assessed by a doctor to determine medical and social

## Hospital Infection Control

### ANTIBIOTIC POLICY

suitability to minimize risk.

#### 1.10 Antibiotic Protocol - ICU

- I. In infections with MDR *Pseudomonas/Acinetobacter*, Carbapenems should be used as Extended Infusions e.g., Imipenem (2-3 hours infusion), Meropenem (3 hours infusion), Doripenem (4 hours infusion)
- II. Linezolid and Daptomycin are reserved drugs for inpatients and should be used only in cases of Vancomycin Resistant *S. aureus* (VRSA) or Vancomycin Resistant *Enterococci* (VRE)
- III. De-escalation to Fluconazole shall be considered only if Isolates are susceptible to Fluconazole (e.g., *Candida albicans*) and patient is clinically stable.
- IV. De-escalation to Voriconazole shall be considered if *C. krusei* or Voriconazole susceptible *C. glabrata* and patient is clinically stable. De-escalation to fluconazole or voriconazole is not recommended without confirmation of isolate susceptibility.
- V. The marker used in our laboratory to assess potential ESBL production among *Enterobacteriaceae* is resistance to Cefotaxime and Ceftazidime.
- VI. The marker used in our laboratory to assess potential MRSA production is the resistance of *S. aureus* to ceftazidime.

#### 1.11 Categorization of antibiotics

The antibiotics shall be categorized by the PTC of the hospital in discussion with IPCC. The categories should be revised at least once in six months.

## Hospital Infection Control

### ANTIBIOTIC POLICY

#### 1.11.1 Restricted use

A pre-use authorization from an ID Physician/Clinical Microbiologist needs to be taken before prescribing these antibiotics. A written documentation to be maintained which captures the request along with justification for use by the clinician and also captures the approval for use by the authority in charge. (Annexure 1: Pre-authorization form)

#### 1.11.2 Limited access

Unrestricted use of these antibiotics may be allowed for empirical use for first 48-72 hours but after that a clinical justification by clinician and approval from authority in charge needs to be documented that why these antibiotics cannot be de-escalated and need to be continued further. (Annexure: Rational antibiotic use monitoring)

#### 1.11.3 Under Surveillance

A close monitoring to check their usage (indication, quantity and pattern) in OPD/Type 1 Patients/ Surgical prophylaxis. Audits shall be done at regular intervals to assess their consumption.

### 1.12 Restricted Use Antibiotics

**1.12.1 Colistin:** It is the last resort for managing gram negative MDRs and its use, dose and duration needs to be rationalized. Liberal use shall be restricted.

**1.12.2 Doripenem:** It is the last Carbapenem (at least in near future). If Imipenem and Meropenem are working, we need to conserve the use of Doripenem.

**1.12.3 Rifampicin:** (For Non-TB use) - This is a valuable drug

## Hospital Infection Control

### ANTIBIOTIC POLICY

for TB. The use of rifampicin in MDR, *Pseudomonas*, *Acinetobacter* or MRSA should be restricted

- 1.12.4 Linezolid: Alternatives available e.g. Vancomycin/Teicoplanin. Linezolid is bacteriostatic and available as oral hence more prone for misuse and VRSA/VRE are rare.
- 1.12.5 Daptomycin: Alternatives are available for MRSA e.g. Vancomycin and Teicoplanin. Moreover, VRSA and VRE are still not a major cause of concern.
- 1.12.6 Tigecycline: Bacteriostatic, one of the most broad-spectrum drugs, has limited role in MDR infections like SSTI, IAI where ESBL/MRSA and/or *Acinetobacter* are feared.
- 1.12.7 Sulbactam: Reserved for PDR *Acinetobacter*. Dose must be correct (4-12 g/day for PDR *Acinetobacter*)
- 1.12.8 Ceftazidime-Avibactam: Reserved for XDR & PDR GNBs proven to be sensitive via E-test/ showing synergy with aztreonam
- 1.12.9 Ceftriaxone-Sulbactam-EDTA: Reserved for XDR & PDR GNBs proven to be sensitive via E-test/ Disc diffusion/ Metallo-beta-lactamase resistance
- 1.12.10 Minocycline: Reserved for CRAB/ *Burkholderia cepacia*/ *Stenotrophomonas matlophilia*
- 1.12.11 Vancomycin/Teicoplanin: Use as empirical in sick patients may be allowed specially in BSI, SSTI where MRSA is suspected but if after 48-72h culture and sensitivity report shows no *S. aureus* or MSSA then Vancomycin/Teicoplanin have absolutely no role and shall be discontinued.
- 1.12.12 Echinocandins: reserved in confirmed candidemia



## Hospital Infection Control

### ANTIBIOTIC POLICY

patients

#### 1.13 Limited Access Antibiotics

**1.13.1** Imipenem/Meropenem: Use as empirical in sick patients is allowed looking at the antibiogram in most hospitals showing better sensitivity of these antibiotics over other classes, however after culture and sensitivity report is available, if it shows a susceptible pathogen to other classes of antibiotics plus if patient condition improves, then de-escalation shall be considered.

**1.13.2** Piperacillin-Tazobactam/Cefoperazone-Sulbactam:

These are as broad spectrum as carbapenems. Use as empirical in sick patients is allowed looking at the antibiogram in most hospitals showing decent sensitivity of these antibiotics over other classes, however after culture and sensitivity report is available, if it shows a susceptible pathogen to other classes of antibiotics plus if patient condition improves, then de-escalation shall be considered.

#### 1.14 Under Surveillance Antibiotics

**1.14.1** 3rd generation Cephalosporins (both oral and IV) and Fluroquinolones: One of the main reasons for widespread ESBLs in India in the community is due to overuse of 3rd generation Cephalosporins and Fluroquinolones at OPD level-Type 1 patients, pediatric patients and surgical prophylaxis. Every opportunity shall be utilized to educate the clinicians about these antibiotics and the collateral damage they cause. Also, it is imperative to exercise control on liberal usage of these antibiotics in a

## Hospital Infection Control

### ANTIBIOTIC POLICY

phased manner and regular audits shall be performed on the rate of consumption of these antibiotics. This could be the single most valuable intervention to curb resistance in India in community.

#### 1.15 Irrational combinations or less evidenced combinations

|             |                                       |              |                                    |
|-------------|---------------------------------------|--------------|------------------------------------|
| <b>I</b>    | Amoxicillin-tazobactam                | <b>XV</b>    | Ceftazidime-tazobactam             |
| <b>II</b>   | Cefadroxil-clavulanic acid            | <b>XVI</b>   | Ceftriaxone-sulbactam              |
| <b>III</b>  | Cefepime + Amikacin                   | <b>XVII</b>  | Ceftriaxone-tazobactam             |
| <b>IV</b>   | Cefepime-sulbactam                    | <b>XVIII</b> | Cefuroxime-clavulanic acid         |
| <b>V</b>    | Cefepime-tazobactam                   | <b>XIX</b>   | Cefuroxime-sulbactam               |
| <b>VI</b>   | Cefixime + Ofloxacin                  | <b>XX</b>    | Meropenem-sulbactam                |
| <b>VII</b>  | Cefixime + Ornidazole                 | <b>XXI</b>   | Vancomycin + Ceftriaxone           |
| <b>VIII</b> | Cefixime-clavulanic acid              | <b>XXII</b>  | Cefoperazone-Tazobactam            |
| <b>IX</b>   | Cefotaxime-sulbactam                  | <b>XXIII</b> | Ampicillin-Amoxicillin-Cloxacillin |
| <b>X</b>    | Cefpodoxime-clavulanic                | <b>XXIV</b>  | Ceftazidime-Sulbactam              |
| <b>XI</b>   | Ofloxacin-Ornidazole/Tinidazole       | <b>XXV</b>   | Doxycycline-Tinidazole             |
| <b>XII</b>  | Gatifloxacin-Ornidazole               | <b>XXVI</b>  | Tetracycline-Metronidazole         |
| <b>XIII</b> | Ciprofloxacin/Gatifloxacin + Ambroxol | <b>XXVII</b> | Roxithromycin+ Ambroxol            |

## Hospital Infection Control

### ANTIBIOTIC POLICY

|            |                        |               |                                                   |
|------------|------------------------|---------------|---------------------------------------------------|
| <b>XIV</b> | Fluconazole-Tinidazole | <b>XXVIII</b> | Cefixime/Cefadroxil +<br>Ambroxol + Lactobacillus |
|------------|------------------------|---------------|---------------------------------------------------|

#### 1.16 Peri-surgical Antibiotic Use

##### 1.16.1 Aim of surgical antibiotic prophylaxis is to:

- I. Prevent surgical site infection (SSI) & related morbidity and mortality
- II. Reduce the duration and cost of health care (when the costs associated with the management of SSI are considered, the cost effectiveness of prophylaxis become evident)
- III. Antibiotic chosen should not produce any adverse effects as well as no adverse consequences for the microbial flora of the patient or the hospital.

Surgical antibiotic prophylaxis is an adjunct to, not a substitute for good surgical technique. Antibiotic prophylaxis should be regarded as one component of an effective policy for the control of healthcare associated infection such as attention to basic infection control strategies, the surgeon's experience and technique, the duration of the procedure,

hospital and operating-room environments, instrument sterilization issues, preoperative preparation (e.g., surgical scrub, skin antisepsis, appropriate hair removal), perioperative management (temperature and glycemic control) and the underlying medical condition of the patient.

##### 1.16.2 Peri-surgical Antibiotic prophylaxis – General principles

- I. Do not use antibiotic prophylaxis routinely for clean, non-prosthetic, uncomplicated surgery.

## Hospital Infection Control

### ANTIBIOTIC POLICY

- I. Give antibiotic treatment (in addition to prophylaxis) to patients having surgery on a dirty or infected wound.
- II. No prophylaxis for class Ia patient except
  - a. Abdominal cases
  - b. Surgery exceeding 2h
  - c. Having three concomitant diagnosis
- III. No prophylaxis for urological procedures with sterile urine
- IV. Give antibiotic prophylaxis to patients before Clean surgery involving the placement of a prosthesis or implant, Clean-contaminated surgery Contaminated surgery.
- V. Antibiotics should also be administered immediately after unexpected contamination of the tissues.
- VI. Consider giving a SINGLE DOSE of antibiotic prophylaxis intravenously within
 

1 hour before incision to maximize tissue concentration.

However, give prophylaxis earlier for operations in which a tourniquet is used. Two hours are allowed for the administration of vancomycin and fluoroquinolones.
- VII. Before giving antibiotic prophylaxis, consider the timing and pharmacokinetics (for example, the serum half-life) and necessary infusion time of the antibiotic. Give a repeat dose of antibiotic prophylaxis at one to two half-lives of the antibiotic when the operation is longer than the half-life of the antibiotic given or if there is excessive blood loss (usually more than 1500 ml in adults) during the procedure or extensive burns.
- VIII. Therapeutic antibiotics to be given for all class III and class IV wounds.
- IX. Post-operative antibiotic administration is NOT required.
- X. There is no data to support the continuation of

## Hospital Infection Control

### ANTIBIOTIC POLICY

antimicrobial prophylaxis until all indwelling drains and intravascular catheters are removed.

- XI. Antibiotic prophylaxis should be confined to the perioperative period (less than 24 hours for most procedures). The prophylaxis duration in cardiothoracic procedures may be up to 48 hours. Prolonged prophylaxis is associated with an increased risk of acquired antimicrobial resistance.
- XII. Discontinue antibiotics given for implantation of a pacemaker or defibrillator within 24 hours of surgery.
- XIII. Select appropriate agents based on the surgical procedure, the most common pathogens causing SSI for a specific procedure and published recommendations.
- XIV. Inform patients before the operation, whenever possible, if they will need antibiotic prophylaxis, and afterwards if they have been given antibiotics during their operation.
- XV. Studies with results showing a beneficial effect of supplemental oxygen included patients who underwent colorectal surgery. It has been observed that 30%-35% supplemental FiO<sub>2</sub> levels are useful in minimizing SSI. Higher/lower concentrations are less helpful.
- XVI. Maintaining normothermia (temperature higher than 36°C) immediately after colorectal surgery is helpful in reducing the incidence of SSI.

### 1.16.3 Choice of antibiotic for Peri-surgical prophylaxis

- I. Antibiotic selection is influenced by the organism most commonly causing wound infection in the specific procedure, characteristics of the ideal agent, the comparative efficacy of the antimicrobial agent for the procedure, the safety profile, and the patient's

## Hospital Infection Control

### ANTIBIOTIC POLICY

medication allergies and cost of the antibiotic agent. In certain gastrointestinal procedures, oral and intravenous administration of agents with activity against Gram-negative and anaerobic bacteria is warranted, as well as mechanical preparation of the bowel. Cefazolin provides adequate coverage for most types of procedures.

- II. Antimicrobial agents with the narrowest spectrum of activity are required for efficacy in preventing infection and the chosen antibiotics must reflect local, disease-specific information about the common pathogens and their antimicrobial susceptibility. A history of a serious adverse event should preclude administration of a particular antibiotic like penicillin.
- III. Do not routinely use vancomycin prophylaxis for any procedure.

### 1.17 Surgical Prophylaxis Guidelines

1. Antibiotics are repeated if the duration of operation is > 4 hours or if blood loss is > 1 liter (except vancomycin, aminoglycoside, fluoroquinolone).
2. Prophylactic antibiotics should be administered within 1 hour prior to incision.
3. Prophylactic antibiotics should not be used in perianal procedures (lay open fistula, hemorrhoidectomy, lateral anal sphincterotomy)
4. Avoid hypothermia as it increases incidence of surgical site infection.
5. No role for prophylactic antibiotic in routine catheterization

#### CDC SURGICAL WOUND CLASSIFICATION:

## Hospital Infection Control

### ANTIBIOTIC POLICY

|                                |                                                              |                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. CLEAN</b>                | <b>1a.</b> Clean operation without use of prosthetic implant | Uninfected, no inflammation<br>Respiratory, Gastrointestinal and Genitourinary tracts not entered<br>Closed primarily<br>The evidence for use of antibiotic is thin.                                                                                                                                                                                 |
|                                | <b>1b.</b> Clean operation with use of prosthetic implant    |                                                                                                                                                                                                                                                                                                                                                      |
| <b>2. CLEAN - CONTAMINATED</b> | <b>2a.</b> Clean contaminated operation                      | Respiratory, Gastrointestinal and Genitourinary tracts entered, controlled, no spillage<br>No unusual contamination.                                                                                                                                                                                                                                 |
|                                | <b>2b.</b> Clean contaminated operation                      | Head & neck operation where oral cavity/upper aero digestive tract is open, including esophageal operations.                                                                                                                                                                                                                                         |
| <b>3. CONTAMINATED</b>         | Contaminated operation                                       | Open, fresh, accidental wounds<br>Major break in sterile technique<br>Gross Spillage from GI tract<br>Acute non-purulent inflammation                                                                                                                                                                                                                |
| <b>4. DIRTY - INFECTED</b>     | Dirty                                                        | Old traumatic wounds, devitalized tissue<br>Existing infection or perforation<br>Organisms present before procedure                                                                                                                                                                                                                                  |
|                                |                                                              | Antibiotics are not “prophylactic” here. Choice of antibiotics will depend on whether organ dysfunction is present or not. Specimens for culture and sensitivity should be taken at operation. If organ dysfunction is present broad-spectrum antibiotics will be chosen initially and de-escalate once culture / sensitivity results are available. |

## Hospital Infection Control

### ANTIBIOTIC POLICY

#### 1.17.1 Dose of antibiotics

Please see table below for dosing and re-dosing guidelines. It is recommended to have weight-based dosing of both cefazolin and vancomycin. Cefazolin should be administered every 4 hours; clindamycin every 8 hours; vancomycin does not require re-dosing given its long half-life. It is also recommended to consider re-dosing earlier than specified in Table if there is excessive intra-operative blood loss (e.g. >1500 mL). However, aminoglycosides and vancomycin should not be re-dosed in this setting.

#### 1.17.2 Timing of the pre-operative antibiotic dose

It is recommended that pre-operative antibiotics shall be administered <60 minutes prior to incision. Pre-operative antibiotics should reach acceptable tissue concentrations prior to the incision time to be effective.

#### 1.17.3 Duration of post-operative antibiotics

It is recommended that all patients receive <24 hours of post-operative antibiotics. In many procedures, no doses after incision closure are necessary.

#### 1.17.4 Choice of antibiotics

Please see table below for acceptable choices of antibiotics based upon surgical procedure. Consider the addition of vancomycin or clindamycin for patients known to be colonized with MRSA. Clindamycin and vancomycin are recommended alternative agents to cefazolin for patients with beta-lactam allergies.



## Hospital Infection Control

### ANTIBIOTIC POLICY

| Nature of Operation                                                           | Etiology                                           | Recommended antimicrobials                                                                                             | Usual adult dose(<120 Kg)                            | Redose Interval                                                     |
|-------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------------------|
| Procedure involving entry into lumen of gastrointestinal tract including ERCP | Enteric gram-negative bacilli, gram-positive cocci | Cefoperazone<br>Sulbactam<br><br><b>OR</b><br><br>Cefotaxime/Ceftriaxone +<br>Amikacin<br><b>PLUS</b><br>Metronidazole | 2g IV<br><br><br><br>1g IV + 5mg/kg<br><br><br>500mg | BD for 24 hrs maximum<br><br><br><br>4 hours<br><br><br>IV infusion |

**\*HIGH RISK - Morbid obesity, gastrointestinal obstruction, decreased gastric acidity or GI motility, gastric bleeding, malignancy or perforation, or immunosuppression.**

#### BILIARY TRACT SURGERY (INCLUDING PANCREATIC)

| Nature of Operation                                  | Etiology                                               | Recommended antimicrobials                             | Usual adult dose                |         |
|------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|---------------------------------|---------|
| Laparoscopic procedure (low risk)                    | N/A                                                    | N/A                                                    | N/A                             | N/A     |
| Open procedure or laparoscopic procedure (high risk) | Enteric gram-negative bacilli, enterococci, clostridia | Cefoperazone<br>Sulbactam<br><br>Cefotaxime + Amikacin | 2g IV<br><br><br>1g IV + 5mg/kg | 4 hours |

**Hospital Infection Control**
**ANTIBIOTIC POLICY**

|                           |                                |                                                 |                            |         |
|---------------------------|--------------------------------|-------------------------------------------------|----------------------------|---------|
| Obstructed biliary system | Enteric gram-negative bacteria | Cefoperazone<br>Sulbactam<br>+<br>Metronidazole | 2 g/1g IV<br><br>500 mg IV | 4 hours |
|---------------------------|--------------------------------|-------------------------------------------------|----------------------------|---------|

**High risk laparoscopic - Age >70 years, acute cholecystitis, non-functioning gall bladder, or common bile duct stones.**

**VAN 1g IV infusion if MRSA high risk**

**COLORECTAL SURGERY**

| Nature of Operation | Etiology                                              | Recommended antimicrobials                      | Usual adult dose       | Redose Interval    |
|---------------------|-------------------------------------------------------|-------------------------------------------------|------------------------|--------------------|
| Colorectal surgery  | Enteric gram-negative bacilli, anaerobes, enterococci | Cefoperazone<br>Sulbactam<br>+<br>Metronidazole | 2 g IV<br><br>500mg IV | 4 hours<br><br>N/A |

**OTHER GENERAL SURGERY**

| Nature of Operation | Etiology | Recommended antimicrobials | Usual adult dose | Redose Interval |
|---------------------|----------|----------------------------|------------------|-----------------|
|---------------------|----------|----------------------------|------------------|-----------------|

**Hospital Infection Control**
**ANTIBIOTIC POLICY**

|                                |                                 |                                                               |                                     |         |
|--------------------------------|---------------------------------|---------------------------------------------------------------|-------------------------------------|---------|
| Hernia repair                  | Aerobic gram-positive organisms | Ampicillin OR Cloxacillin OR Cefotaxime/Ceftriaxone           | 1 g IV                              | 4 hours |
| Minor Breast Procedure (Clean) |                                 |                                                               | 1 g IV                              |         |
| Radical Mastectomy             | Aerobic gram-positive organisms | Cefoperazone<br>Sulbactam<br>OR<br>Cefuroxime<br>Or Cefazolin | 2 g IV<br><br>1.5gm IV<br><br>2g IV | 4 hours |

**HEAD AND NECK SURGERY**

| Nature of Operation                                                                                                                                                                                                                 | Etiology                                            | Recommended antimicrobials                                       | Usual adult dose                           | Redose Interval    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|------------------------------------------------------------------|--------------------------------------------|--------------------|
| Clean<br><br>including thyroidectomy                                                                                                                                                                                                | -                                                   | None                                                             | N/A                                        | N/A                |
| Clean - Contaminated<br>Tonsillectomy*,<br>adenoidectomy*,<br>uncomplicated ear surgery, otoplasty, stapedectomy, nasal septoplasty, endoscopic sinus surgery, other uncomplicated nose or sinus surgery and minor clean procedures | Anaerobes, enteric gram-negative bacilli, S. aureus | Cefazolin/Cefotaxime<br>+<br>Metronidazole                       | 2 g / 1 g IV<br><br>500mg IV               | 4 hours<br><br>N/A |
| Major ear surgery<br>Revision sinus surgery<br>Complex septorhinoplasty<br>Tympanomastoid surgery                                                                                                                                   | Anaerobes, enteric gram-negative bacilli, S. aureus | cefazolin IV<br>30mg/kg/dose (up to 2g) PLUS<br>metronidazole IV | vancomycin IV<br>30mg/kg/dose (up to 1.5g) | 4 hours<br><br>N/A |

## Hospital Infection Control

### ANTIBIOTIC POLICY

|                                    |  |                                                                                                                                                                                      |                                                                                                                                                             |  |
|------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Laryngectomy (primary or salvage)* |  | 12.5mg/kg/dose (up to 500mg) High risk of MRSA infection: ADD vancomycin IV 30mg/kg/dose (up to 1.5g) *POSTOPERATIVE doses can be considered but should not continue beyond 24 hours | PLUS (for laryngectomy or tympanomastoid surgery) gentamicin IV 2mg/kg/dose^ *POSTOPERATIVE doses can be considered but should not continue beyond 24 hours |  |
|------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

### ORAL / DENTAL SURGERY

|                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                          |     |                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Clean or clean-contaminated procedures not listed below Dental extractions, impactions, exposures, implants, minor pathology (soft tissue, cysts)                                | Prophylaxis recommended                                                                                                                                                                                                                                                                                                  | NOT | Endocarditis prophylaxis                                                                                                                           |
| Insertion of prosthetic material (except dental implants) Intraoral bone grafting Open reduction and internal fixation of mandibular or midfacial fractures Orthognathic surgery | cefazolin IV 30mg/kg/dose (up to 2g) PLUS (if insertion through the skin and oral mucosa) metronidazole IV 12.5mg/kg/dose (up to 500mg) High risk of MRSA infection: ADD vancomycin IV 30mg/kg/dose (up to 1.5g) *POSTOPERATIVE doses can be considered for orthognathic surgery but should not continue beyond 24 hours |     | clindamycin IV 15mg/kg/dose (up to 600mg) *POSTOPERATIVE doses can be considered for orthognathic surgery but should not continue beyond 24 hours. |

### VASCULAR SURGERY

| Nature of Operation | Etiology | Recommended antimicrobials | Usual adult dose | Redose Interval |
|---------------------|----------|----------------------------|------------------|-----------------|
|---------------------|----------|----------------------------|------------------|-----------------|

## Hospital Infection Control

### ANTIBIOTIC POLICY

|                                                                          |                           |                                                                                           |                              |         |
|--------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------|------------------------------|---------|
| Arterial surgery involving prosthesis, abdominal aorta or groin incision | S. aureus, S. epidermidis | Cefazolin<br><b>OR</b><br>Cefuroxime<br><b>ADD</b> if contaminated major wounds, Amikacin | 2 g IV<br>1.5 g IV<br>500 mg | 4 hours |
| Superficial venous system surgery                                        | S. aureus, S. epidermidis | N/A                                                                                       | N/A                          | N/A     |

### ORTHOPEDIC SURGERY

| Nature of Operation                                                                       | Etiology                        | Recommended antimicrobials                                                      | Usual adult dose | Redose Interval |
|-------------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------|------------------|-----------------|
| Clean surgeries including fracture repair, arthroplasty, implantation of foreign material | Aerobic gram-positive organisms | Cefotaxime/<br>Cefuroxime/Ceftriaxone<br><b>OR</b><br>Cefoperazone<br>Sulbactam | 2 g IV           | 4 hours         |

**Hospital Infection Control**
**ANTIBIOTIC POLICY**

|                             |                                            |                                                                                |                         |                   |
|-----------------------------|--------------------------------------------|--------------------------------------------------------------------------------|-------------------------|-------------------|
| Clean Contaminated or Dirty | Aerobic gram-positive organisms, Anaerobes | Cefoperazone<br>Sulbactam<br><b>PLUS</b><br>Ornidazole                         | 2 g IV<br><br>300mg IV  | 4 hours<br><br>NA |
| Joint replacement           | Aerobic gram-positive organisms            | Cefoperazone<br>Sulbactam<br><b>ADD</b> In case of MRSA High Risk, Teicoplanin | 2 g IV<br><br>400 mg IV | 4 hours<br><br>NA |

**NEUROSURGEY/CTVS**

| Nature of Operation                      | Etiology            | Recommended antimicrobials | Usual adult dose | Redose Interval |
|------------------------------------------|---------------------|----------------------------|------------------|-----------------|
| Neurosurgery                             | -                   | Cefoperazone<br>Sulbactam  | 2 g IV           | 4 hours         |
| Cardiac Surgery/<br>Vascular/Thoracic    | -                   | Cefazolin/Cefotaxime       | 2 g IV           | 4 hours         |
| Cardiac Surgery with prosthetic material | Gram-positive cocci | Cefazolin/Cefotaxime<br>+  | 2 g IV           | 4 hours         |

## Hospital Infection Control

### ANTIBIOTIC POLICY

|                                                                  |                              |                      |                                                                  |         |
|------------------------------------------------------------------|------------------------------|----------------------|------------------------------------------------------------------|---------|
|                                                                  |                              | Vancomycin           | < 80 kg = 1g<br>80-99 kg=1.25g<br>100-120 kg =1.5g<br>>120 kg=2g |         |
| Cardiac device insertion<br>(e.g.,<br>pacemaker<br>implantation) | S. aureus,<br>S. epidermidis | Cefazolin/Cefotaxime | 2 g IV                                                           | 4 hours |

### GENITO-UROSURGERY

| Nature of Operation                                                                                                                                  | Etiology                                 | Recommended antimicrobials                                                                                              | Usual adult dose       | Re dose Interval |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------|------------------|
| Urological Procedures when pre-op urine culture data is unavailable, or cultures were negative. UDS, TRPB, DJ stent removal, TURP, TURBT, URSL, ESWL | Enteric gram-negative bacilli            | Cefazolin/Cefotaxime<br>If prosthetic materials involved in urologic procedures, should add one-time dose of Gentamicin | 1 g IV<br><br>80 mg IV | 4 hours          |
| Urological procedures like PCNL, Radical/Partial Nephrectomy, Adrenalectomy, Radical Prostatectomy                                                   | Enteric gram-negative bacilli, S. aureus | Cefoperazone<br>Sulbactam IV                                                                                            | 2 g IV                 | 4 hours          |
| Cesarean delivery                                                                                                                                    | Gram-negative bacilli, S. aureus         | Cefazolin<br><b>OR</b><br>Cefuroxime                                                                                    | 2 g IV<br><br>1.5 g IV | 4 hours          |

## Hospital Infection Control

### ANTIBIOTIC POLICY

|                                       |                                                 |                                                                      |                |         |
|---------------------------------------|-------------------------------------------------|----------------------------------------------------------------------|----------------|---------|
| Gynecological<br>(eg<br>hysterectomy) | Enteric gram-<br>negative bacilli, S.<br>aureus | Cefazolin/Cefuroxim<br>e<br><br>To add a dose<br>of<br>Metronidazole | 2 g / 1.5 g IV | 4 hours |
|---------------------------------------|-------------------------------------------------|----------------------------------------------------------------------|----------------|---------|

## Surgical Antibiotic Prophylaxis - Pediatric

| Surgery                                   | No penicillin allergy                                                                                                                                                                                                                                                         | Alternatives                                                                                                                                                                                                                                                                  |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Head & Neck (ENT - Clean)                 | Cefazolin 30 mg/kg IV (maximum 2,000 mg for patients < 120 kg, and 3,000 mg for patients ≥ 120 kg)                                                                                                                                                                            | Vancomycin 15 mg/kg IV (max 2,000 mg)                                                                                                                                                                                                                                         |
| Head & Neck (ENT – Clean Contaminated)    | Ampicillin/sulbactam 50 mg/kg IV (maximum 2,000 mg; dose based on ampicillin component)                                                                                                                                                                                       | Vancomycin 15 mg/kg IV (max 2,000 mg) and • Metronidazole 15 mg/kg IV (max 500 mg)                                                                                                                                                                                            |
| General Surgery (Upper Gastrointestinal1) | Cefazolin 30 mg/kg IV (maximum 2,000 mg for patients < 120 kg, and 3,000 mg for patients ≥ 120 kg)                                                                                                                                                                            | Vancomycin 15 mg/kg IV (max 2,000 mg)                                                                                                                                                                                                                                         |
| General Surgery (Lower Gastrointestinal2) | <ul style="list-style-type: none"> <li>Cefoxitin 40 mg/kg IV (maximum 2,000 mg) preferred regimen or</li> <li>Cefazolin 30 mg/kg IV (maximum 2,000 mg for patients &lt; 120 kg, and 3,000 mg for patients ≥ 120 kg) and metronidazole 15 mg/kg IV (maximum 500 mg)</li> </ul> | <ul style="list-style-type: none"> <li>Cefoxitin 40 mg/kg IV (maximum 2,000 mg) preferred regimen or</li> <li>Cefazolin 30 mg/kg IV (maximum 2,000 mg for patients &lt; 120 kg, and 3,000 mg for patients ≥ 120 kg) and metronidazole 15 mg/kg IV (maximum 500 mg)</li> </ul> |
| Orthopedic                                | Cefazolin 30 mg/kg IV (maximum 2,000 mg for patients < 120 kg, and 3,000 mg for patients ≥ 120 kg)                                                                                                                                                                            | Vancomycin 15 mg/kg IV (max 2,000 mg)                                                                                                                                                                                                                                         |

### 1.1 Wound classification algorithm:

Did you encounter:

Purulence/existing clinical infection?  
Perforated viscera? Open traumatic wounds > 4 hours? Retained devitalized tissue? Penetrating injuries > 4 hours?

YES

Class IV

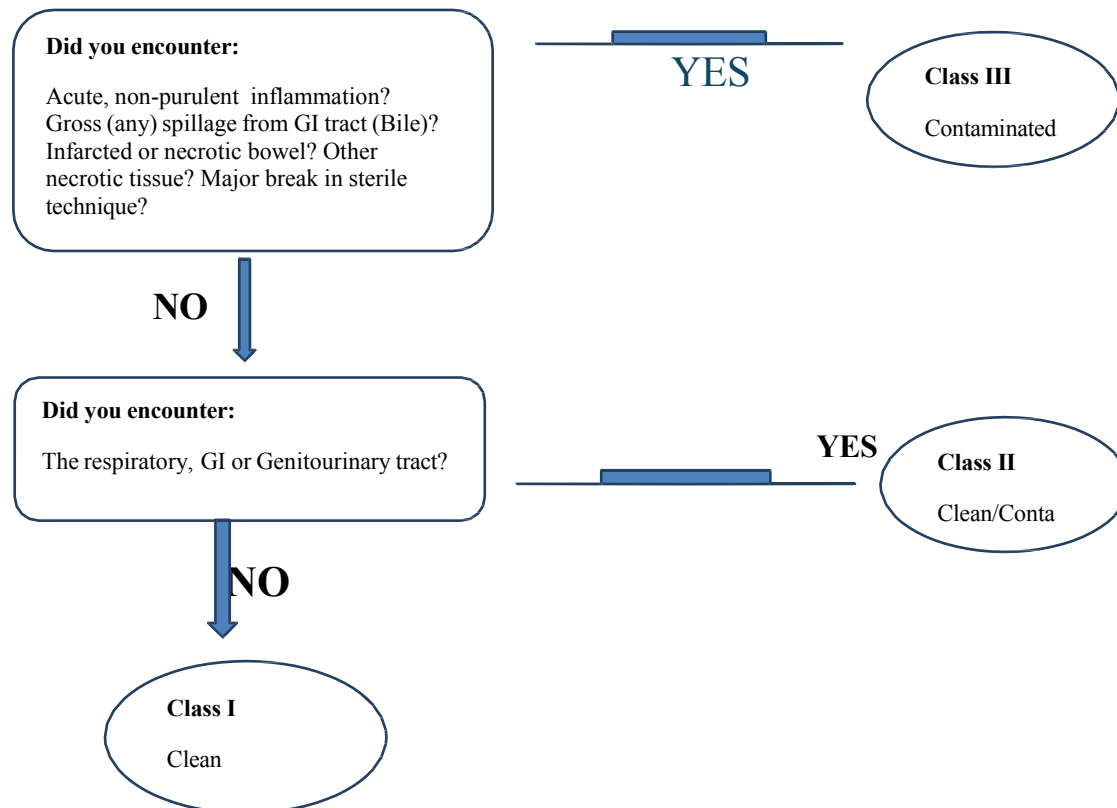
Dirty/Infect

NO



## Hospital Infection Control

### ANTIBIOTIC POLICY



## 2. Empiric treatment for common infections

### 2.1 GASTRO - INTESTINAL INFECTIONS

| Condition | Most likely<br>Microbial<br>etiology | 1st Choice | Alternatives | Comments |
|-----------|--------------------------------------|------------|--------------|----------|
|           |                                      |            |              |          |

## Hospital Infection Control

### ANTIBIOTIC POLICY

|                                                                       |                                                                                                 |                                                                                                                 |                                                                           |                                                                                              |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Acute gastroenteritis (acute onset nausea, vomiting, watery diarrhea) | Viral (caliciviruses, rotaviruses), Enterotoxigenic & Enteropathogenic E. coli, Salmonella spp. | None indicated                                                                                                  | None                                                                      | <ul style="list-style-type: none"> <li>Rehydration</li> <li>Symptomatic treatment</li> </ul> |
| Acute watery diarrhea, cholera suspected                              | Vibrio cholerae                                                                                 | Doxycycline 300mg PO 1 dose<br><br><b>OR</b><br>Azithromycin Oral in children (20mg/kg) and pregnant women (1g) | Azithromycin 1gm PO 1 dose <b>OR</b><br>Ciprofloxacin 500mg PO bid 3 days | Prompt rehydration essential<br><br>Antibiotics are adjunctive therapy.                      |

|                                                             |                                                                        |                                                               |                                                                           |                    |
|-------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------|--------------------|
| Bacillary dysentery (acute onset fever and bloody diarrhea) | Campylobacter spp.<br>Shigella spp.<br><br>Non-typhoidal Salmonellosis | None needed for previously healthy patient with mild symptoms | Ciprofloxacin 500mg PO bid 3 days<br><br>Azithromycin 500mg PO OD 3 doses | Prompt rehydration |
|-------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------|--------------------|

**Hospital Infection Control**
**ANTIBIOTIC POLICY**

|                                                                   |                                             |                                                                                                       |                                   |                                                                          |
|-------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------|--------------------------------------------------------------------------|
|                                                                   | Shiga toxin producing E. coli               | Treat patients with severe symptoms immunocompromised status<br>Antibiotic Treatment not recommended. |                                   | Antibiotic use associated with development of hemolytic uremic syndrome. |
| Amoebic dysentery                                                 | E. histolytica                              | Metronidazole 400mg Oral TDS for 7-10 days                                                            | Tinidazole 2gm Oral OD for 3 days | Add diloxanide furoate 500 mg TDS for 10d                                |
| Enteric fever – suspect if AFI ≥ 7 days, other etiology ruled out | Salmonella typhi<br>Salmonella Para typhi A | Antibiotic treatment should be based on culture & susceptibility reports<br><u>Outpatients:</u>       |                                   | Majority of strains                                                      |

## Hospital Infection Control

### ANTIBIOTIC POLICY

|  |  |                                                                                                                                                                                                            |                                          |                                                                                                                                                                 |
|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | Cefixime<br>20mg/kg/day<br>for 14 days or<br>Azithromycin<br>500mg BD for 7<br>days.<br><br><u>Inpatients:</u><br>Ceftriaxone 2<br>g<br>IV BD for 2<br>weeks +/-<br>Azithromycin<br>500mg BD for 7<br>days | Cotrimoxazol<br>960 mg BD<br>for 2 weeks | are Nalidixic<br>acid resistant.<br><br>Ceftriaxone to be<br>changed to oral<br>Cefixime when<br>patient is afebrile to<br>finish total duration of<br>14 days. |
|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                        |                                 |                                                                                                                                       |                                          |                                                                         |
|------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------|
| Cholangitis            | Enterobacteriaceae<br>Anaerobes | Piperacillin-<br>Tazobactam<br>4.5g IV Q8H<br>Ertapenem 1 g<br>IV OD<br>(for severely<br>ill patients –<br>sepsis or septic<br>shock) | Cefoperazone-<br>Sulbactam<br>3 g IV BID | Duration: 7<br>days<br>Biliary<br>drainage                              |
| Acute<br>cholecystitis | Enterobacteriaceae              | Piperacillin-<br>Tazobactam 4.5g<br>IV Q8H<br>Ertapenem 1 g                                                                           | Cefoperazone-<br>Sulbactam 3 g<br>IV BID | Patients undergoing<br>cholecystectomy<br>should have<br>antimicrobials |

**Hospital Infection Control**
**ANTIBIOTIC POLICY**

|                                                                                        |                                                                                 |                                                                                                                    |                                                                                             |                                                                                                                                                              |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                        |                                                                                 | IV<br>OD<br><br>(for severely ill patients – sepsis or septic shock)                                               |                                                                                             | discontinued within 24 h unless there is evidence of infection outside the wall of the gallbladder                                                           |
| Spontaneous bacterial peritonitis                                                      | E. coli                                                                         | Piperacillin-Tazobactam 4.5g IV Q8H<br>Ertapenem 1 g IV OD<br>(for severely ill patients – sepsis or septic shock) | Cefoperazone-Sulbactam 3 g IV BID                                                           | Duration: 7 days                                                                                                                                             |
| Secondary peritonitis (bowel perforation)                                              | Enterobacteriaceae (E.coli, Klebsiella sp.),<br>Anaerobes (Bacteroides species) | Ertapenem 1 g IV OD                                                                                                | Cefoperazone-Sulbactam 3 g IV BID                                                           | Emergency surgery to eliminate source of contamination, reduce bacterial load & prevent recurrence Duration: 5 - 7 days; longer if source control inadequate |
| Intra-abdominal abscess                                                                | Enterobacteriaceae<br>Anaerobes (Bacteroides spp)                               | Ertapenem 1 g IV OD                                                                                                | Cefoperazone-Sulbactam 3 g IV BID<br>Tigecycline 100mg IV x 1 dose, followed by 50mg IV BID | Emergency drainage Duration: 5 - 7 days; longer if source control inadequate                                                                                 |
| Amoebic liver abscess (suspect in patients with single abscess in right lobe of liver) | E. histolytica                                                                  | Metronidazole 500 mg IV TID or 800 mg PO TID<br>+ Diloxanide                                                       |                                                                                             | Therapeutic drainage for: high risk of abscess rupture, left lobe liver abscess                                                                              |

## Hospital Infection Control

### ANTIBIOTIC POLICY

|                                                                   |                                            |                                                                                    |                                                    |                                                                                                |
|-------------------------------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------|
| with no<br>IHBRD and no<br>primary intra-<br>abdominal<br>source) |                                            | furoate 500<br>mg TID x 10<br>days                                                 |                                                    |                                                                                                |
| Giardiasis                                                        | Giardia lamblia                            | Metronidazole<br>250- 500mg<br>oral<br>TID x 7-10 d                                | Tinidazol<br>e2 gm<br>oral x1<br>dose              |                                                                                                |
| Diverticulit<br>is Mild-<br>OPD treatment                         | Gram-<br>Negative<br>Bacteria<br>Anaerobes | Amoxycillin-<br>Clavulanat<br>e625mg<br>TDS for 7 days                             | Ciprofloxacin<br>+<br>Metronidazol<br>e for 7 days |                                                                                                |
| Diverticulit<br>is moderate                                       | Gram-<br>Negative<br>Bacteria<br>Anaerobes | Ceftriaxone<br>2gm IV<br>OD<br>+metronidazole<br>500 mg IV TDS<br>or Piperacillin- |                                                    | BL-BLI agents<br><br>have very good<br>anaerobic cover,<br>so no need to add<br>metronidazole. |

**Hospital Infection Control**
**ANTIBIOTIC POLICY**

|                       |                                                            |                                                                                                                     |  |                                                                                                                                                              |
|-----------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       |                                                            | Tazobactam<br>4.5gm IV<br><br>8 hourly<br>empirically<br>or<br><br>Cefoperazone-<br>Sulbactam<br>3gm IV 8<br>hourly |  |                                                                                                                                                              |
| Diverticulitis Severe | Gram-Negative<br>Bacteria<br>Anaerobes                     | Meropenem<br>1gm IV<br><br>8hrly or<br>Imipenem<br>Cilastatin<br>500mg IV<br>6<br>hourly                            |  | Duration based on improvement                                                                                                                                |
| Acute pancreatitis    | Routine use of prophylactic antibiotics<br>NOT recommended |                                                                                                                     |  | Infected pancreatic necrosis should be considered in patients who deteriorate or fail to improve after 10 days of hospitalization or CT scan with gas in the |

**pancreas**

In these patients, either CT-guided FNA for Gram stain and culture to guide use of appropriate antibiotics or Empiric antibiotics (e.g., Meropenem 1 g IV TID) may be given.

- Ref: ACG Guidelines 2013

## Hospital Infection Control

### ANTIBIOTIC POLICY

**\*When using a Carbapenem (e.g. ertapenem or meropenem) or beta-lactam + beta-lactamase inhibitor combination (e.g. piperacillin-tazobactam) for intra-abdominal infections, there is NO NEED to add metronidazole. Carbapenems and BI + BLI combinations have excellent activity against anaerobes.**

## 2.2 GENITO-URINARY INFECTIONS

### UTI management protocol

#### 1. Symptoms and signs

- a. Young women: dysuria, frequency of urination, and negative history of vaginal discharge (post-test probability 90%); fever; costo-vertebral angle tenderness
- b. Older patients may have these additional symptoms
  - i. Cloudy, malodorous or bloody urine
  - ii. Worsening incontinence
  - iii. Systemic symptoms (altered mental status, MOSF, hypotension)
- c. Symptoms of urinary catheter associated UTI (CAUTI):  
new onset or worsening of fever, rigors, altered mental status, malaise, or lethargy with no other identified cause;  
flank pain; costo-vertebral angle tenderness; acute hematuria; pelvic discomfort

#### 2. Lab tests

- a. Urinalysis (leukocyte esterase and nitrite by dipstick).  
Absence of pyuria and bacteriuria rules out UTI (high negative predictive value)
- b. CBC
- c. U & E
- d. Urine culture
- e. Blood culture
- f. Imaging (ultrasound or CT KUB) indicated for
  - i. Severe sepsis or septic shock



## Hospital Infection Control

### ANTIBIOTIC POLICY

- ii. Palpable kidneys
- iii. Persistent symptoms after 72 hrs. appropriate treatment
- iv. DM
- v. Immune suppression
- vi. Recurrent UTI

| Condition                                                                                             | Most likely Microbial etiology | 1st Choice              | Alternatives                                                     | Comments                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Asymptomatic bacteriuria (positive urine culture from an individual without symptoms or signs of UTI) | E. coli                        | No antimicrobial needed | Can consider Amoxicillin 500mg PO TID if indicated as mentioned. | Screening for and treatment of asymptomatic bacteriuria is indicated for <ol style="list-style-type: none"> <li>1. Pregnant women</li> <li>2. Patients undergoing</li> <li>3. Urologic procedures in which mucosal bleeding is anticipated</li> </ol> |

## Hospital Infection Control

### ANTIBIOTIC POLICY

|                                                                                                                                                                           |                             |                                                                                                                                    |                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acute uncomplicated cystitis in women – dysuria and frequency in healthy, adult, non-pregnant women with normal urinary tract (no structural or functional abnormalities) | E. coli, Enterobacteriaceae | Nitrofurantoin 100 mg PO BID x 7 days                                                                                              | Co-trimoxazole, Amoxicillin, Ampicillin, Cephalexin                            | <b>Pregnancy -</b> Cephalexin or nitrofurantoin for 7 days / short course (3 days) co- trimoxazole (avoid during last trimester).<br><b>Relapse-</b> No treatment for non-pregnant women without obstructive uropathy or symptoms of UTI.<br><b>Relapse- treat for 2 weeks, Second relapse- treat for 4-6 weeks, further relapses – treat longer periods.</b><br><b>Relapse during pregnancy- obtain follow- up cultures monthly.</b><br><b>For men- evaluate for chronic bacterial prostatitis.</b> |
| Pyelonephritis – uncomplicated (no underlying GU disease)                                                                                                                 | E. coli Enterobacteriaceae  | Mild to moderate illness: Piperacillin-Tazobactam 4.5 g IV TID<br><br>Severe illness (sepsis or septic shock): Ertapenem 1 g IV OD | Cefoperazone, Sulbactam, Amikacin<br><br>Consider oral step down after 7 days. | <b>Definitive therapy based on culture and susceptibility report.</b><br><b>Duration:</b><br><b>Mild to moderate cases - 7 days,</b><br><b>Severe cases - 14 days</b>                                                                                                                                                                                                                                                                                                                                |

## Hospital Infection Control

### ANTIBIOTIC POLICY

|                                                                                                      |                                                              |                      |                                                                                |                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Complicated UTI (underlying GU disease, e.g., neurogenic bladder, renal stones, hydronephrosis etc.) | E. coli, Proteus, Pseudomonas aeruginosa                     | Meropenem 1 g IV TID | Consider adding amikacin or colistin in seriously ill (physician's discretion) | <ul style="list-style-type: none"> <li>• <b>Definitive therapy based on culture and susceptibility report</b></li> <li>• <b>Duration: 10 - 14 days</b></li> </ul> |
| Foley's catheter associated UTI                                                                      | E. coli, Proteus, Pseudomonas aeruginosa, Acinetobacter spp. | No empiric treatment |                                                                                | <b>Definitive therapy based on culture and susceptibility report</b>                                                                                              |

## Hospital Infection Control

### ANTIBIOTIC POLICY

|                                                   |                                                                         |                                                                                                                                               |                                                  |                                                                 |
|---------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------|
| Acute prostatitis, uncomplicated, low risk of STD | Enterobacteriae, Enterococcus                                           | Cipro or levofloxacin/ co-trimoxazole for 10-14 days                                                                                          | In drug resistance- use Carbapenem, upto 4 weeks | <b>Risk of STD - Consider gonococci-abscess needs drainage.</b> |
| Pelvic inflammatory disease                       | N. gonorrhoea, C. trachomatis, Anaerobes, GNB, Streptococci, Mycoplasma | Mild to mod- OP - Ceftriaxone 250mg IM or IV (1 dose) + Doxycycline +/- Metronidazole for 14 days IP - Clindamycin + Gentamycin + Doxycycline | IP - Cefaperazone- Sulbactam + Doxycycline       |                                                                 |

## Hospital Infection Control

### ANTIBIOTIC POLICY

|                                                  |                       |                                                                                                      |  |                                                                                                                                            |
|--------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------|
| Clostridium difficile associated diarrhea (CDAD) | Clostridium difficile | Mild -<br>Metronidazole 400mg PO TID for 10 days<br>Moderate -<br>Vancomycin 250mg PO QID for 10 day |  | <b>Discontinue culprit antibiotic</b><br><br><b>In severe (toxic megacolon) – Vancomycin retention enema with caution, consult surgeon</b> |
|--------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------|

Do not send urine culture from an asymptomatic patient with indwelling urinary catheter

Urinalysis for pyuria NOT useful in diagnosing CAUTI

Treat only when patient has symptoms attributable to UTI

Urine sample for culture should be obtained either through a new catheter (after removing the indwelling catheter) or through sample port near junction of drainage tubing and Foley catheter. Do not send samples from the drainage bag. Remove urinary catheter if not indicated.

Replace urinary catheter only if essential; consider alternatives (e.g. condom catheter, intermittent catheterization etc.)

Duration of antimicrobials: 7 – 14 days

Candiduria - if catheter associated, removal of catheter- cures in 30-40%. Oral Fluconazole 200- 400mg/day for 7 days, along with removal (if possible) or replacing the catheter (may be effective for only short term). Ascending infections in diabetics or obstruction, urologic procedures- need treatment even if Non-catheterized.

## 2.3 NEUROLOGICAL INFECTIONS

### I. Management protocol for acute meningitis (community acquired)

#### 1. Suspect if patient (no neurosurgical procedure in the previous

## Hospital Infection Control

### ANTIBIOTIC POLICY

two weeks; not immuno-compromised) has any combination of the following:

- a. Symptoms: fever, headache, altered mental status  
(new onset confusion, disorientation, drowsiness, coma)
- b. Signs: Temp  $>38^{\circ}\text{C}$ , neck stiffness, other signs of meningeal irritation.

## 2. Confirm by a lumbar puncture (to be done as soon as possible)

- a. **CSF findings** highly suggestive of bacterial meningitis
  - i. Gross appearance turbid
  - ii. Total WBC count  $>1000$  cells/mm<sup>3</sup>
  - iii. Neutrophilic pleocytosis
  - iv. Low CSF glucose ( $<50\%$  of concomitant blood glucose)

## 3. CT scan brain

- a. Should NOT be ordered as a routine test before LP in all cases of suspected acute meningitis.
- b. Indications for CT scan brain before LP:
  - i. Papilledema
  - ii. New onset seizures
  - iii. Focal neurological deficits (e.g. hemiparesis)
  - iv. Decreased level of consciousness (GCS  $<10$ )
  - v. Immunocompromised patients (e.g. HIV infection)

## 4. Lab tests

- a. CSF analysis
  - i. WBC count (total and differential)
  - ii. Glucose
  - iii. Protein
  - iv. Gram stain and culture
  - v. Virology panel (PCR)
  - vi. Additional investigations if duration of symptoms  $>5$  days
    1. Xpert MTB/RIF PCR

## Hospital Infection Control

### ANTIBIOTIC POLICY

2. Mycobacterial (MGIT) culture
  3. India ink preparation
  4. Cryptococcal antigen
  5. Fungal culture
  6. VDRL
- b. Blood culture
  - c. CBC
  - d. Urea, creatinine, electrolytes

## 5. Antimicrobial management:

- a. All patients should receive the first dose of antimicrobials as soon as the diagnosis of acute bacterial meningitis is suspected.
- b. DO NOT delay antimicrobials if there is a delay in obtaining a CSF sample.
  - Prior administration of antimicrobials tends to have minimal effects on the chemistry and cytology findings, but can reduce the yield of Gram stain and culture.

## C. Initial empiric antimicrobials

- i. Ceftriaxone 2 g IV BID +/- Vancomycin 10–20 mg/kg q8 - 12h to achieve serum trough concentrations of 15–20 mg/ml. Alternatives are Meropenem +/- Vancomycin.
- ii. Consider Listeria in elderly, pregnancy, immunocompromised, add ampicillin 2gQID
- iii. Modify antimicrobial regimen based on results of culture & susceptibility reports.
- iv. Duration: 10 – 14 days
- v. Adjunctive steroid therapy to be started for patients with strong clinical suspicion of acute bacterial meningitis or CSF results as described below:

## Hospital Infection Control

### ANTIBIOTIC POLICY

#### 1. Indications:

- Cloudy CSF
- CSF WBC count >1000/ml, or
- Bacteria in CSF on Gram's staining

#### B. Contra-indications

- Septic shock
- Patients who have already received antimicrobials

#### C. Dose and duration: Dexamethasone 0.15 mg/kg

Q6HX 4 days; first dose 15min before or along with

first dose of antimicrobial

#### D. Discontinue if culture grows organisms other than *Strep pneumoniae*

## II. Other CNS infections

| Condition                                        | Most likely Microbialetiology | Preferred Choice       | Alternative s                     | Comments                                                                     |
|--------------------------------------------------|-------------------------------|------------------------|-----------------------------------|------------------------------------------------------------------------------|
| Septic cavernoussinus thrombosis                 | S. aureus                     | Cloxacillin 2 g IVQ4H  |                                   | Duration: 3 - 4weeks                                                         |
| Meningitis, postneuro-surgical, penetrating head | Staphylococci, GNB            | Meropenem + Vancomycin | cefepime/ ceftazidime +vancomycin | Revise based onculture report.<br>In shunt infections - shuntmust be revised |



## Hospital Infection Control

### ANTIBIOTIC POLICY

|                                    |                                                                             |                                                   |                                         |                                                                      |
|------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------|-----------------------------------------|----------------------------------------------------------------------|
| Brain abscess (community acquired) | Staphylococcus, Streptococcus, Enterobacteriaceae, Anaerobes                | Ceftriaxone + vancomycin + metronidazole          |                                         | Surgical drainage<br><br>Treat as per culture & sensitivity reports. |
| Brain abscess (neurosurgical)      | Staphylococcus, Streptococcus, Enterobacteriaceae, Anaerobes, P. aeruginosa | Cefepime/Ceftazidime + Vancomycin + Metronidazole | Meropenem /Piptazo + vancomycin         | Surgical drainage                                                    |
| Epidural abscess                   | Staphylococcus, Streptococcus, anaerobes, GNB HSV, Varicella zoster         | Vancomycin + cefepime/ceftazidime                 | May step down when culture is available | Surgical drainage-4-6 weeks of antibiotic                            |
| Viral encephalitis                 | HSV, Varicella zoster                                                       | IV Acyclovir 10mg/kg Q8H for 10-14 days           |                                         |                                                                      |

## 2.4 ORTHOPEDIC INFECTIONS

**Hospital Infection Control**
**ANTIBIOTIC POLICY**

| Condition                                                                                                                                                                                                                                                         | Most likely<br>Microbial<br>etiology                | 1st Choice                                                                          | Alternatives                | Comments                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acute osteomyelitis                                                                                                                                                                                                                                               | S. aureus                                           | Cefazolin 2 g<br>IV TID                                                             | Cefalexin<br>500mg PO<br>BD | <ul style="list-style-type: none"> <li>For optimal treatment, microbial etiology should be confirmed</li> <li>Consider bone biopsy and debridement of necrotic material</li> <li>Modify initial empiric regimen based on culture report</li> <li>Duration: 6 weeks from last debridement</li> </ul> |
| Chronic osteomyelitis <ul style="list-style-type: none"> <li>Secondary to a contiguous focus of infection (e.g. decubitus ulcer)</li> <li>Osteomyelitis that develops because of contaminated open fractures or surgical treatment of closed fractures</li> </ul> | Staphylococci, Aerobic GNB, Streptococci, Anaerobes | Avoid empiric treatment,<br>Consider Clindamycin 600mg TID + Ciprofloxacin 500mg BD |                             | <ul style="list-style-type: none"> <li>For optimal treatment, microbial etiology should be confirmed</li> <li>Consider bone biopsy and debridement of necrotic material</li> <li>Obtain cultures (bone and blood) before antimicrobials.</li> </ul>                                                 |

## Hospital Infection Control

### ANTIBIOTIC POLICY

|                                                       |                                         |                                                           |                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                       |                                         |                                                           | <ul style="list-style-type: none"> <li>• Avoid sending swab cultures from chronic discharging sinuses and ulcers</li> <li>• Duration: 6 weeks from last debridement</li> </ul>                                                                                                                                        |
| Chronic osteomyelitis with orthopedic implants        |                                         | Avoid empiric antimicrobials unless patient seriously ill | <ul style="list-style-type: none"> <li>• For optimal treatment, microbial etiology should be confirmed</li> <li>• Obtain cultures (bone and blood) before antimicrobials</li> <li>• Modify initial empiric regimen based on culture report</li> <li>• Implant removal and debridement of necrotic material</li> </ul> |
| Osteomyelitis associated with diabetic foot infection | See section on diabetic foot infections |                                                           |                                                                                                                                                                                                                                                                                                                       |

**Hospital Infection Control**
**ANTIBIOTIC POLICY**

|                                               |               |                                                                                                                 |                         |                                                                                                                                                                                                       |
|-----------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Septic arthritis                              | S. aureus     | Cloxacillin 2 g<br>IV Q6H<br><br>If MRSA<br>suspected<br>consider<br>Vancomycin<br>and stepdown<br>to Linezolid | Cefazolin<br>2 g IV TID | Obtain joint fluid for culture<br>before starting<br>antimicrobials<br>Modify initial empiric<br>regimen based on culture<br>report<br>Duration: 4 weeks; change<br>to PO after 2 weeks IV<br>therapy |
| Septic arthritis with suspected<br>Gonococcus | N. gonorrhoea | Ceftriaxone 1g<br>IV BD                                                                                         |                         | Look for h/o contact,<br>urethritis                                                                                                                                                                   |

## 2.5 RESPIRATORY INFECTIONS

### 1. Community acquired pneumonia (CAP) management protocol

**1.1 Suspect** if patient (not been hospitalized in the previous 90 days; not immuno- compromised; no structural lung disease like bronchiectasis) has any combination of the following:

- I. Symptoms: fever, cough (with or without expectoration), shortness of breath, chestpain
- II. Signs: Temp >38°C, tachypnea, tachycardia, impaired percussion notes, bronchial breath sounds, crackles, altered VF/VR.
- III. In a patient presenting with cough, normal vital signs and physical examination findings rule out a diagnosis of pneumonia

**1.2 Confirm** with chest x-ray (to be done as soon as possible). Inpatients with a clinical suspicion of CAP,

## Hospital Infection Control

### ANTIBIOTIC POLICY

but no abnormalities on chest x-ray, repeat the x-ray within 48 hours.

#### 1.3 Severity assessment based on CURB 65 score: 6-point score (range 0 - 5)

1.3.1 Gives one point each for:

- I. Confusion (new onset disorientation in person, place or time)
- II. Urea >126 mg/dL
- III. Respiratory rate > 30/min
- IV. Low Blood pressure (SBP < 90 mm Hg or DBP < 60 mm Hg)
- V. Age >65 years

1.3.2 **Interpretation:** CURB 65 score is NOT a replacement for good clinical judgment.

- I. CURB-65 score 0 or 1: low risk of death
- II. CURB-65 score 2: moderate risk of death
- III. CURB-65 score >3: high risk of death

#### 1.4 Lab tests

- I. CBC
- II. Urea, creatinine, electrolytes
- III. For patients with CURB 65 score >2
  - a. ABG
  - b. Blood culture
  - c. Sputum Gram stain & culture (Sputum sample should be transported promptly to the lab)
  - d. Respiratory sample (NP swab, throat swab, nasal swab, ET aspirate or BAL) for respiratory viral panel (includes influenza testing by RT-PCR). In patients with pneumonia, BAL or ET aspirate should be collected for influenza testing if NP swab is negative

## Hospital Infection Control

### ANTIBIOTIC POLICY

## 1.5 Setting of care

- I. CURB-65 score 0 or 1: Out-patient
- II. CURB-65 score 2: In-patient (Ward)
- III. CURB-65 score 3: In-patient (ICU)

## 1.6 Antimicrobial management:

- I. All patients should receive the first dose of antimicrobials as soon as the diagnosis of CAP is confirmed, *as per the table below*.
- II. Change to an oral regimen as soon as clinical improvement occurs and the temperature has been normal for 24 h, and there is no contraindication to the oral route.
- III. Modify antimicrobial regimen based on results of culture & susceptibility reports
- IV. Consider unusual microbial etiology (e.g. *Burkholderia pseudomallei* in poorly controlled diabetes and *Staph aureus* (MSSA & MRSA) in post-influenza bacterial pneumonia)

## 2. Ventilator associated pneumonia management protocol

- I. Develops 48 – 72 hours after endo tracheal intubation
- II. Duration of appropriate antimicrobial therapy: 8 – 10 days.
- III. Clinical features: fever, alteration in sputum characteristics (increased purulence and /or volume), worsening oxygenation (increasing FiO<sub>2</sub> & PEEP requirement, worsening PF ratio)
- IV. Labs: leukocytosis (WBC >11000), leukopenia (WBC <4000)

## Hospital Infection Control

### ANTIBIOTIC POLICY

or band forms

>10%, elevated PCT / CRP

- V. Chest x-ray: new or worsening infiltrates
- VI. No gold standard for diagnosis of VAP;  
combination of above findings increases probability of VAP
- VII. Obtain ET aspirate for Gram stain, cultures and virology (No advantage of BAL over ETA)
- VIII. Negative ET aspirate culture rules out VAP (very high negative predictive value)

#### 2.1 Antimicrobial management (to be started after obtaining cultures)

- I. If **hemodynamic instability (systolic BP <90 mm Hg)**  
requiring inotropes: Meropenem 2 g IV TID + Colistin 9 million units IV loading dose, followed by 4.5 million units IV BID
- II. If **no hemodynamic instability**: Meropenem 1 g IV TID + Amikacin 15 mg/kg IV OD
- III. Consider Vancomycin/Linezolid if MRSA incidence high.
- IV. Modify once culture and sensitivity report available

### 3. Influenza-like illness (ILI) protocol

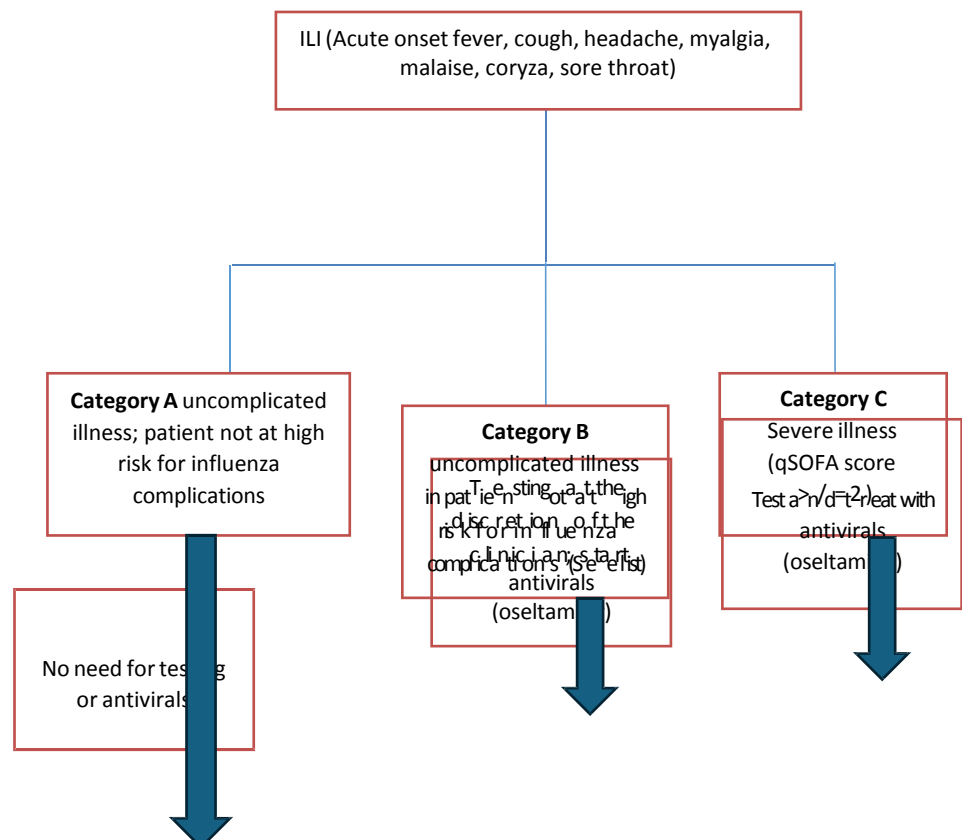
#### 3.1 Patients at higher risk of influenza complications

- I. Children <2 years
- II. Adults >65 years
- III. Pregnant women
- IV. Persons with following comorbidities

Morbid obesity  
COPD, bronchial asthma

CAD, heart failure  
CKD  
CLD  
Hematological conditions (including sickle cell disease)  
DM  
Neurological & neuromuscular disorders  
Immunosuppression (HIV infection, immunosuppressive treatment)

### 3.2 Management Protocol for ILI





**Hospital Infection Control**
**ANTIBIOTIC POLICY**

## 4. Management of Pneumonia as per CURB-65 Score

| CURB-65 score | Most likely microbial etiology                                                                         | Preferred                                                                                                       | Alternate                                                                                           | Comments                                                                                                                                                                                                                                                                                                                                                                             |
|---------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 or 1        | Respiratory viruses<br>Strep pneumoniae<br>Mycoplasma pneumoniae<br>Chlamydia pneumoniae               | Amoxicillin 500 mg<br>PO TID x 5 days                                                                           | Azithromycin 500mg PO OD x 5 days<br>Doxycycline 100 mg PO BID x 5 days<br>Levofloxacin 750mg PO OD | See section on "influenza" for indications for oseltamivir<br>Avoid FQ till PTB is ruled out by sputum examination.<br>Consider FQ when CAP with recent antibiotic exposure.                                                                                                                                                                                                         |
| 2             | Respiratory Virus<br>Strep pneumoniae<br>Mycoplasma pneumoniae<br>Chlamydia pneumoniae<br>Legionella   | Penicillin G 20L IV Q4H + Azithromycin 500 mg IV OD x 5-7 days + Oseltamivir 75 mg PO BID x 5 days              | Ceftriaxone 1 g IV OD + Azithromycin 500 mg IV OD x 5-7 days + Oseltamivir 75 mg PO BID x 5 days    | Antiviral treatment might still be beneficial in patients with severe, complicated or progressive illness and in hospitalized patients when started after 48 hours of illness onset, as indicated by observational studies (CDC, 2013)<br><br>Discontinue oseltamivir if PCR negative,<br>Continue if clinical suspicion of influenza high.<br>No virological or clinical advantages |
| 3             | S.pneumoniae,<br>Legionella,<br>K.pneumoniae,<br>H influenza, Respiratory viruses, primarily influenza | Piperacillin-Tazobactam 4.5 g IV Q8H + Azithromycin 500 mg IV OD x 5-7 days + Oseltamivir 75 mg PO BID x 5 days |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                      |

## 5. Other respiratory infections

| Condition   | Most likely microbial etiology | Preferred             | Alternative | See Comments* |
|-------------|--------------------------------|-----------------------|-------------|---------------|
| Common Cold | Viral                          | Symptomatic treatment |             |               |

## Hospital Infection Control

### ANTIBIOTIC POLICY

|                   |                                              |                                     |                                    |                                                                  |
|-------------------|----------------------------------------------|-------------------------------------|------------------------------------|------------------------------------------------------------------|
| Oral thrush       | Candida spp.                                 | Fluconazole 100-200mg OD for 7 days | Clotrimazole troche                | Check for previous antibiotic use, underlying immunosuppression. |
| Acute pharyngitis | Group A streptococcus<br>Respiratory viruses | Amoxicillin 500 mg PO TID x 10 days | Azithromycin 500 mg PO OD x 5 days |                                                                  |

|                              |                                                                                                                                                               |                                                                                                                                                                                                                                                 |                                                                                                                                               |                                                                                                                                                                                |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Acute epiglottitis           | H. influenza<br><br>Children:<br><br>H.influenzae ,<br>S.pyogenes,<br>S.pneumoniae,<br>S.aureus.<br><br>Adult: Group A<br>Streptococcus ,<br><br>H.influenzae | Ceftriaxone 1-2 g<br>IV OD x 7<br>days                                                                                                                                                                                                          | Cefotaxime 1-2gm IV<br>Q8H                                                                                                                    | Airway management -<br>Intubation if indicated                                                                                                                                 |
| Community acquired Pneumonia | S. pneumoniae,<br><br>H.influenza e,<br>Legionella,<br>E.coli, Klebsiella sp.,<br>S.aureus                                                                    | Mild to moderate cases<br>Amoxicillin-500mg-1 g TDS, Oral.<br>If IV indicated, amoxicillin-clavulanate 1.2 g IV TDS OR Ceftriaxone 2g IV OD<br><br>For Severe cases Amoxicillin-clavulanate g IV TDS OR Ceftriaxone 2g IV OD, Duration 5-8 days | Piperacillin-<br><br>Tazobactam 4.5gm<br><br>IV 6 hourly OR Imipenem 1g IV<br><br>6 hourly OR Cefoperazone-<br><br>Sulbactam 3gm IV 12 hourly | If MRSA is a concern, add Linezolid 600mg IV/Oral BD<br>Vancomycin 15 - 20mg/kg<br><br>If atypical pneumonia suspected, Doxycycline 100mg bd OR Azithromycin 500 mg oral/IV OD |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
|                                                                                                                                    | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
|                                                                                                                                    | Date of Approval:30/04/2024             |
| <b>Hospital Infection Control</b>                                                                                                  | Date of Review:30/05/2025               |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Page <b>59</b> of <b>98</b>             |

|                                    |                         |                                                                                                                                                                                                                                                                                                                                   |                                           |
|------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Hospital acquired Pneumonia</b> | P. aeruginosa           | <b>1. Patient without risk factor for MDR</b><br>Inj. Ceftriaxone 2g IV, Q24H OR<br>Inj. Cefotaxime 2g IV, Q6- 8 H, 8 days or if CPIS decrease                                                                                                                                                                                    |                                           |
|                                    | Staphylococcus aureus   | <b>2. Patient with risk factor for MDR</b><br>1. Beta lactam - Inj Cefazidime 2g IV Q8H<br>OR<br>Inj. Cefepime 2g IV Q8- Q12 H OR<br>Inj. Piperacillin + Tazobactam 4.5g IV Q6H<br>OR<br>Inj. Imipenem, 500 mg IV Q6 H OR 1g IV Q8 OR<br>Inj. Meropenem, 1 g IV Q8H<br>In areas where isolation of gram-negative organism is high |                                           |
|                                    | E.coli, Klebsiella sp., | <b>2. For Gram negative pathogen</b><br>Inj. Gentamicin OR Tobramycin 7mg/kg IV OD<br>OR Inj. Amikacin 20mg/kg IV OD OR<br>Inj. Ciprofloxacin 400 mg IVQ8H OR<br>Inj. Levofloxacin 750mg IV Q24H<br><b>3. For Gram positive</b><br>Inj. Linezolid 600 mg IV                                                                       | Inj. Vancomycin 15mg/kg Q12H later adjust |

|                                                                                                                                    |                                         |  |  |  |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--|--|--|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |  |  |  |
|                                                                                                                                    | Date of Issue:30/06/2024                |  |  |  |
|                                                                                                                                    | Revision Date and Status: 30/05/2024/03 |  |  |  |
|                                                                                                                                    | Date of Creation:20/05/2024             |  |  |  |
|                                                                                                                                    | Date of Approval:30/04/2024             |  |  |  |
| <b>Hospital Infection Control</b>                                                                                                  | Date of Review:30/05/2025               |  |  |  |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Page <b>60</b> of <b>98</b>             |  |  |  |

|  |  |      |  |  |
|--|--|------|--|--|
|  |  | Q12H |  |  |
|--|--|------|--|--|

|                                                                                                                                                    |                                                                                                                |                                                                                   |                                                                                                                                                                                                      |                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Deep neck space infection including Ludwig's angina                                                                                                | H. influenzae<br>Polymicrobial (oral anaerobes)<br>Streptococcus spp.<br>Pepto streptococci<br>Bacteroides spp | Ceftriaxone 1-2 g IV OD x 7 days<br>Amoxicillin-Clavulanate 1.2gm IV BD x 14 days | Clindamycin 600 mg IV. Q8H x 14 days                                                                                                                                                                 | Airway management<br>Surgery<br>Continue antibiotics till clear evidence of clinical improvement |
| Acute bronchiolitis                                                                                                                                | Virus - RSV, Parainfluenza, Adeno                                                                              | None needed                                                                       |                                                                                                                                                                                                      | Ribavirin has questionable role.                                                                 |
| Acute Bronchitis                                                                                                                                   | Mycoplasma pneumonia, Hemophilus influenzae                                                                    | Amoxicillin + Clavulanic acid 625mg TID for 5-7 days                              | Doxycycline 100mg BD for 7 days OR Erythromycin 500mg QID for 5-7 days                                                                                                                               |                                                                                                  |
| Acute laryngitis                                                                                                                                   | Rhino virus, RSV, parainfluenza                                                                                | No antibiotics, voice rest, steam inhalations.                                    | Ampicillin-sulbactam 2gm iv q4h OR Clindamycin IV 600mg TID                                                                                                                                          | Rule out GERD, Penicillin/Amox not shown much clinical benefit                                   |
| Acute otitis media, Acute bacterial rhinosinusitis (Antimicrobials if symptoms of fever, facial pain and purulent nasal discharge persist >7 days) | S. Pneumoniae<br>H. influenza<br>M. catarrhalis                                                                | Amoxicillin 500mg TID x 10 day<br>Amoxicillin + Clavulanic acid 625mg BD          | Amoxicillin-Clavulanate 1 g PO BID x 7 days OR Cefuroxime 250/500mg BD x 5 days OR Azithromycin 500mg OD oral OR Ciprofloxacin 500mg BD oral OR T. clarithromycin 250mg BD OR Roxithromycin 150mg bd | Duration of treatment: 5-7 days                                                                  |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
|                                                                                                                                    | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
|                                                                                                                                    | Date of Approval:30/04/2024             |
| <b>Hospital Infection Control</b>                                                                                                  | Date of Review:30/05/2025               |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Page <b>61</b> of <b>98</b>             |

|                                                                                                                                                                        |                                                                            |                                                                                                             |                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Malignant otitis Externa                                                                                                                                               | pseudomonas                                                                | Inj Levofloxacin 500mg OD, if response is good-change to oral Ciprofloxacin 500mg BD – continue for 6 weeks | Inj Cefoperaz one + Sulbactam 2gm iv BD                                                              | Duration of Treatment: According to the clinical condition and host immunity                                                 |
| Mastoiditis: Acute                                                                                                                                                     | S.pneumoniae<br><br>S.aureus<br>H.influenzae<br>P.aeruginosa               | Cefotaxime 1-2 gm iv 4-8 hourly<br>Ceftriaxone 2 gm iv OD                                                   |                                                                                                      | Modify as per culture Unusual causes- Nocardia, TB, Actinomyces.                                                             |
| Mastoiditis: Chronic                                                                                                                                                   | Polymicrobial                                                              | Piperacillin-tazobactam 4.5g, IV 8h OR Meropenem 1 gm iv 8h                                                 |                                                                                                      |                                                                                                                              |
| Aspergillosis                                                                                                                                                          |                                                                            | Invasive: Voriconazole 6 mg/ kg IV Q 12 hourly followed by 4 mg/ kg IV every 12 hourly                      | Liposomal Amphotericin B 3- 5 mg/ kg IV daily or Caspofungin 70 mg IV loading followed by 50 mg/ day | Duration of therapy up to 6 - 12 weeks                                                                                       |
| Allergic Broncho Pulmonary Aspergillosis (ABPA)                                                                                                                        |                                                                            | Steroids are the main stay of the therapy                                                                   | Voriconazole 400 mg OD                                                                               | Duration of therapy: upto 16 weeks                                                                                           |
|                                                                                                                                                                        |                                                                            | Itraconazole 200 - 400 mg/ day for 16 weeks                                                                 |                                                                                                      | Recurrent are common there for may needed longer anti-fungal courses.                                                        |
| Acute bacterial exacerbation of COPD (presence of at least 2 of the following 3 symptoms:<br>Increased dyspnea, increased sputum volume and increased sputum purulence | Strep. Pneumoniae<br>H. influenza<br>M. catarrhalis<br>Respiratory viruses | Amoxicillin-Clavulanate 1 g PO BID x 7 days                                                                 | Azithromycin 500 mg PO OD x 3 days OR Doxycycline 100 mg PO BID x 7 days                             | Start oseltamivir if clinical suspicion of influenza high Antimicrobials for all patients who require mechanical ventilation |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
|                                                                                                                                    | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
|                                                                                                                                    | Date of Approval:30/04/2024             |
| <b>Hospital Infection Control</b>                                                                                                  | Date of Review:30/05/2025               |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Page 62 of 98                           |

|                                                                                   |                                                                                                          |                                                                                                                                                 |                                                                              |                                                                                                   |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Bronchiectasis, acute exacerbation (increased cough, sputum volume and purulence) | H. influenzae<br>S. pneumoniae<br>P. aeruginosa                                                          | Initial empiric therapy based on previous sputum culture reports when available If no report available:<br>Piperacillin-Tazobactam 4.5 g IV TID |                                                                              | Duration of treatment individualized. Postural drainage.                                          |
| Lung abscess                                                                      | Oral anaerobes (Peptostreptococcus, Prevotella, Bacteroides (usually not B. fragilis) Fusobacterium spp. | Amoxicillin-Clavulanate 1.2 g IV TID                                                                                                            |                                                                              | Duration: Till clinical and radiological resolution; usually 3 to 4 weeks                         |
| Acute Tonsillitis                                                                 | Viral Group A Streptococci                                                                               | Amoxicillin + Clavulanic acid 375-625mg TID oral                                                                                                | Azithromycin 500mg OD oral<br>Cap Clindamycin 300mg BD                       | Duration of treatment: 7-10 days                                                                  |
| Empyema thoracis                                                                  | Strep. milleri<br>Strep. Pneumoniae<br>Anaerobes                                                         | As per culture reports                                                                                                                          | Clindamycin 600 mg IV TID,<br>Amox-Clav 1.2 g IVTID, Penicillin G 20L IV Q4H | Modify antibiotics once culture reports available<br>Intercostal tube drainage when pleural fluid |

## 2.6 SEPSIS & OTHER BLOOD STREAM INFECTIONS

- Sepsis is a life-threatening organ dysfunction caused by host response to infection. Septicshock is a subset of sepsis with shock, cellular and metabolic abnormalities which is associated with an increased risk of mortality.
- Suspect sepsis** when some of the following are present:
  - General: fever ( $>38^{\circ}\text{C}$ ), hypothermia ( $< 36^{\circ}\text{C}$ ), tachycardia ( $\text{HR}>90$ ), tachypnea ( $\text{RR}>20$ ), altered mental status.
  - Inflammatory: leukocytosis, leukopenia,  $>10\%$  band forms, thrombocytopenia (consumption), elevated plasma C-reactive protein, elevated procalcitonin.

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue: 30/06/2024               |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation: 20/05/2024            |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval: 30/04/2024            |
|                                                                                                                                    | Date of Review: 30/05/2025              |
| Page 63 of 98                                                                                                                      |                                         |

- c. Hemodynamic: arterial hypotension (SBP < 90 mm Hg, MAP < 70 mm Hg, or SBP decrease > 40 mm Hg)
- d. Organ dysfunction: arterial hypoxemia, acute oliguria, creatinine increase > 0.5 mg/dL, coagulation abnormalities (INR > 1.5 or aPTT > 60 sec), ileus, platelets < 1 lakh, hyperbilirubinemia (bilirubin > 4 mg/dL)
- e. Decreased tissue perfusion: elevated lactate > 1 mmol/L, decreased capillary refill or mottling

### 3. Prompt management of hypotension and hypoxia as per the emergency and critical care management protocol

- 4. Look for common source of sepsis: urinary tract infections, intra-abdominal sources, pneumonia, skin and soft tissue infection and vascular line.
- 5. Assess possibility of hospital acquired infection

### 6. Obtain cultures:

- a. Blood cultures 10ml x 2 sets before initiating antibiotics (one drawn percutaneously and the other through central line if present)
- b. Other relevant cultures based on the suspected source (e.g. urine, sputum, pus etc.)
- c. Imaging studies to evaluate the source and to determine need for source control.

### 7. Common pathogens causing sepsis:

- a. Gram-negative bacteria: *E. coli*, *Klebsiella*, *Enterobacter*, *Pseudomonas*, *Acinetobacter*
- b. Gram-positive bacteria: *Staphylococcus aureus*, *S. pneumoniae*, *Streptococcus spp.*
- c. Toxin mediated: Streptococcal or Staphylococcal toxic shock syndrome

### 8. Antimicrobial management:

**A. All patients should receive the first dose of antimicrobials as soon as possible, preferably within 1 hour of presentation.**

B. Empiric antibiotic selection should be based on the source of infection and the local susceptibility profile of the suspected organism causing the infection.

C. Initial empiric antimicrobials for sepsis when the source is unclear:

- i. **Community acquired:** Meropenem 1-2 g IV TID

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
|                                                                                                                                    | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
| <b>Hospital Infection Control</b>                                                                                                  | Date of Approval:30/04/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Review:30/05/2025               |
|                                                                                                                                    | Page <b>64</b> of <b>98</b>             |

- ii. **Hospital acquired infection with septic shock:** Meropenem 2 g  
 IV TID +Colistin 9 million units IV loading, followed by 4.5 million  
 units IV BID

D. Add clindamycin when toxic shock is suspected and  
 vancomycin whenMRSA is likely.

E. Modify antimicrobial regimen in 48-72 hours based on results of culture  
 & susceptibility reports, the site of infection and the clinical status of  
 the patient.

#### F. Candidemia (Candida isolated from a single peripheral or central-line bloodculture):

- i. Start treatment with Caspofungin 70 mg IV x one dose, followed by  
 50 mg IVOD  
 Fluconazole 400 mg IV OD is appropriate for patient with no features of  
 sepsisor septic shock.
- ii. Obtain blood cultures on alternate days until negative to document  
 clearance of Candidemia.
- iii. Ophthalmology evaluation (fundoscopic examination) to r/o retinal involvement
- iv. Echocardiography for patients with persistently positive (>96 hours  
 after antifungal therapy) or clinical features suggestive of IE
- v. Central venous catheters should be removed as early as possible when  
 presumed to be the source and the catheter can be removed safely;  
 decision should be individualized for each patient
- vi. Duration of treatment: 14 days from end of Candidemia (for patients  
 without obvious metastatic complications), change to Fluconazole 400  
 mg PO OD after 5 to 7 days if isolate susceptible and patient clinically  
 stable

#### G. *Staphylococcus aureus* bacteremia (SAB)

- i. Detailed history and physical examination to:
  - 1. Determine source of bacteremia
  - 2. Ascertain presence of prosthetic devices



|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue: 30/06/2024               |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation: 20/05/2024            |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval: 30/04/2024            |
|                                                                                                                                    | Date of Review: 30/05/2025              |
| Page 65 of 98                                                                                                                      |                                         |

3. Evidence of metastatic infection (e.g. endocarditis, bone and joint infection, splenic abscess, renal abscess etc.)
- ii. Echocardiography required for all patients with SAB to r/o endocarditis.  
TTE may be adequate for patients without identified risk factors for IE
  1. Indications for TEE: persistently positive (>96 hours after appropriate antimicrobials) blood cultures, presence of prosthetic heart valve, peripheral stigmata of IE, cardiac valvular abnormalities, metastatic infections.
- iii. Obtain blood cultures on alternate days to document clearance of bacteremia
- iv. Antimicrobials:
  1. MRSA: Vancomycin 15 mg/kg IV BID
  2. MSSA: Cloxacillin 2 g IV Q4H (alternative Cefazolin 2 g IV Q8H)
- v. Duration: 14 days from end of bacteremia for uncomplicated illness; longer duration for IE, metastatic infections, delayed clearance of bacteremia.

## 2.7 Skin and soft tissue infections Management protocol for necrotizing fasciitis (NF)

1. **Suspect NF** in any patient presenting with SSTI who has severe pain, severe sepsis or septic shock
2. **Symptoms and signs:**
  - a. Pain (out of proportion to physical findings), swelling, redness and warmth of affected area
  - b. Changes in skin color from red-purple to patches of blue-gray
  - c. Bullae containing thick pink or purple fluid, crepitus
  - d. Cutaneous gangrene
  - e. Compartment syndrome
  - f. MOSF
  - g. Hypotension

|                                                                                   |                                         |
|-----------------------------------------------------------------------------------|-----------------------------------------|
|  | Doc No: KASPP/AB/01                     |
|                                                                                   | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                 | Revision Date and Status: 30/05/2024/03 |
|                                                                                   | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                          | Date of Approval:30/04/2024             |
|                                                                                   | Date of Review:30/05/2025               |
| Page 66 of 98                                                                     |                                         |

3. Obtain **urgent surgical consultation**. Surgical exploration is the only way to definitively establish the diagnosis of necrotizing infection.
4. Labs
  - a. CBC
  - b. Urea, creatinine, electrolytes
  - c. CPK
  - d. Blood culture
  - e. Gram stain and culture of tissue and pus sample obtained at surgery
5. **Imaging** (plain x-rays, CT scan, MRI scan) to look for presence of gas. The presence of gas in the fascial planes is a highly specific finding, but not very sensitive. **DO NOT delay surgical intervention**.

### 6. Management

- a. Surgery: Aggressive debridement of all necrotic tissue
- b. Antimicrobials:
  - i. Initial empiric treatment: Meropenem 1 g IV TID + Clindamycin 900 mg IV TID
  - ii. Modify antimicrobials according to culture and sensitivity report
  - iii. For **confirmed group A streptococcus infection**: Penicillin G 2 million units IVQ4H + Clindamycin 900 mg IV TID

| Other Skin & Soft Tissue Infections |                                |            |              |          |
|-------------------------------------|--------------------------------|------------|--------------|----------|
| Condition                           | Most likely Microbial etiology | 1st Choice | Alternatives | Comments |
| A. Skin infections                  |                                |            |              |          |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
|                                                                                                                                    | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
|                                                                                                                                    | Date of Approval:30/04/2024             |
| <b>Hospital Infection Control</b>                                                                                                  | Date of Review:30/05/2025               |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Page 67 of 98                           |

|                                                            |                                     |                                                                                                                     |                                           |                                                                                                                          |
|------------------------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Cellulitis (no purulent drainage)                          | Strep.<br>Pyogenes<br><br>S. aureus | Cefazolin 1 g IV Q8H x 7-10 days<br>(for hospitalized patients with moderate illness)<br><br>Cloxacillin 1 g IV Q6H | Cloxacillin 500 mg PO Q6H x 7-10 days     | Oral therapy for milder illness and step-down following improvement with IV therapy<br><br>Duration: until clinical cure |
| Furuncles, carbuncles, cutaneous abscesses (purulent SSTI) | Staph aureus                        | Cloxacillin 500 mg Q6H PO x 7-10 days<br><br>Or Azithromycin 500mg OD 5 days                                        | Topical - Mupirocin cream                 | Incision & drainage                                                                                                      |
| Bacterial vaginosis                                        |                                     | Metronidazole 400 mg BD 7 days                                                                                      | Clindamycin 300mg BD 7 days               |                                                                                                                          |
| Trichomoniasis                                             |                                     | T. Metronidazole 400mg, BD 7 days                                                                                   |                                           |                                                                                                                          |
| Candida vaginitis                                          |                                     | T. Fluconazole 150mg single dose                                                                                    | Clotrimazole 1% cream intravaginal 7 days |                                                                                                                          |
|                                                            |                                     |                                                                                                                     | OR                                        |                                                                                                                          |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
|                                                                                                                                    | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
|                                                                                                                                    | Date of Approval:30/04/2024             |
| <b>Hospital Infection Control</b>                                                                                                  | Date of Review:30/05/2025               |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Page <b>68</b> of <b>98</b>             |

|                        |  |                                                                                                        |                                                                                                                              |  |
|------------------------|--|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|--|
|                        |  |                                                                                                        | Miconazole 2% cream<br>intravaginal 7 days                                                                                   |  |
| Chlamydial Infections  |  | Doxycycline<br>100mg, BD 7days                                                                         | T. Azithromycin,<br>1gm single dose                                                                                          |  |
| Gonococcal Infections  |  | T. Cefixime,<br>400mg, Single dose                                                                     | T. Azithromycin<br>1gm, Single dose                                                                                          |  |
| Genital herpes simplex |  | First episode<br><br>T. Acyclovir<br>400mg, TID, 7-10 days OR<br>T. Valacyclovir<br>1gm, BID 7-10 days | Episodic treatment<br><br><b>a)</b> Acyclovir<br>400mg, TID<br>5 days<br>OR<br><b>b)</b> Valacyclovir<br>500mg BID<br>3 days |  |
|                        |  |                                                                                                        | Suppressive Treatment (More than 6 episodes per year)                                                                        |  |
|                        |  |                                                                                                        | c) T. Acyclovir<br>400mg BID/ day                                                                                            |  |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval:30/04/2024             |
|                                                                                                                                    | Date of Review:30/05/2025               |
| Page 69 of 98                                                                                                                      |                                         |

|                                    |                   |                                                                                                             |                                                                       |                     |
|------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------|
|                                    |                   |                                                                                                             | d)Valacyclovir<br>500mg, OD/day                                       |                     |
| Non-gonococcal urethritis          |                   | T. Azithromycin<br>1g, Single dose                                                                          | C. Doxycycline<br>100mg, BID<br>7 days                                |                     |
| Syphilis                           |                   | Primary/Secondary/early latent (<one year):<br>Inj. Benzathine penicillin 2.4 million units IM, single dose | For Penicillin Allergic Patients<br>- Doxycycline<br>100mg BD 2 Weeks |                     |
|                                    |                   | Latent (> one year)<br>Inj. Benzathine penicillin<br>2.4million units<br>IM - weekly * 3 weeks              | Doxycycline<br>100mg, BD * 4 weeks                                    |                     |
| Viral infections<br>Herpes simplex |                   | T. Acyclovir, 400 mg, TID, 7 days                                                                           | Valacyclovir 1g, BID, 7 days                                          |                     |
| Tinea versicolor                   | Malassezia furfur | Topical Clotrimazole 1%<br>OD<br>T. Fluconazole 400mg, single                                               | Topical miconazole 2%<br>OD<br>T. Fluconazole 150mg weekly            | Duration 1- 2 weeks |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval:30/04/2024             |
|                                                                                                                                    | Date of Review:30/05/2025               |

Page 70 of 98

|                |           |                                                                                 |                                                                                     |                     |
|----------------|-----------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------|
|                |           | dose                                                                            | once * 4-6 weeks                                                                    |                     |
| Tinea corporis | T. rubrum | Topical<br>Clotrimazole, 1%<br>OD                                               | Topical<br>miconazole 2%<br>OD                                                      | Duration 2- 4 weeks |
|                |           | T. Fluconazole<br>150mg, once<br>weekly, 4-6<br>weeks                           | T. Terbinafine<br>(<35 kg) -125mg<br>OD, 2 weeks.<br>(>35kg) - 250mg<br>OD, 2 weeks |                     |
|                |           |                                                                                 | C. Itraconazole<br>100mg, OD<br>2weeks 200mg,<br>OD, 1<br>weeks                     |                     |
|                |           |                                                                                 | C. Griseofulvin<br>500mg (10-<br>20mg/kg/day)<br>OD 2 weeks                         |                     |
| Tinea capitis  |           | T. Terbinafine<br><br>(<35kg) 125mg 4-<br>6weeks<br>(>35kg) 250kg 4-<br>6 weeks |                                                                                     |                     |

|                                                                                                                                   |                                         |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                   | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                 | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                   | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                          | Date of Approval:30/04/2024             |
|                                                                                                                                   | Date of Review:30/05/2025               |
|                                                                                                                                   | Page 71 of 98                           |

|               |                   |                                                                                                                          |                                                                                                                     |                                                                                            |
|---------------|-------------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Onychomycosis |                   | Terbinafine, 200 mg OD for 6 - 12 weeks OR Itraconazole, 200 mg OD for 6 - 12 weeks Both can be given as Pulse Therapies | Fluconazole 150 mg per week for more than 2-4 weeks<br>Topical: Clotrimazole lotion, Amorolfine lacquer once a week |                                                                                            |
| Scabies       | Sarcoptes scabiei | Permethrin cream 5% to be applied all over the body from neck downwards                                                  |                                                                                                                     | Wash off after 8-14 hours<br>Clothing should be washed in hot water and dried before reuse |

| <b>B. Diabetic foot infections</b>                                                                   |                                |                     |                                              |                                                                 |
|------------------------------------------------------------------------------------------------------|--------------------------------|---------------------|----------------------------------------------|-----------------------------------------------------------------|
| Mild (local infectioninvolving only skin and subcutaneous tissue, erythema <2 cm, no systemic signs) | Staph aureus<br>Strep pyogenes | Cefazolin 1 g IVQ8H | Cloxacillin 500 - 1000 mg PO Q6H x 7-10 days | Cultures should not betaken from clinically non-infected wounds |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
|                                                                                                                                    | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
|                                                                                                                                    | Date of Approval:30/04/2024             |
| <b>Hospital Infection Control</b>                                                                                                  | Date of Review:30/05/2025               |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Page <b>72</b> of <b>98</b>             |

|                                                                                                                                                                           |                                                                                       |                                                                  |                                                        |                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------|
| Moderate (local infection with erythema >2 cm, or involving structures deeper than skin and subcutaneous tissues(e.g. abscess, osteomyelitis, septicarthritis, fasciitis) | Polymicrobial - Staph aureus, Group A Strep, aerobic Gram-negative bacilli, Anaerobes | Piperacillin-Tazobactam 4.5 gIV Q8H + Vancomycin 15mg/kg IV Q12H |                                                        | Surgical consultation for drainage or debridement |
| Severe (limb threatening - severe cellulitis /gangrene /severe sepsis /septic shock)                                                                                      | Polymicrobial - Staph aureus, Group A Strep, aerobic Gram-negative bacilli, anaerobes | Meropenem 1 gIV TID + Vancomycin 15mg/kg IV Q12H                 | Tigecycline 100mg IV x 1 dose followed by 50 mg IV BID | Surgical consultation for drainage or debridement |

### C. VZV infections

|            |                                                                                                                          |                                                                                                        |                                               |                                                                                                                                                           |
|------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chickenpox | Varicella zoster virus, In case of pregnancy: >20 wks of gestation, presenting within 24 hours of the onset of the rash: | Valacyclovir 1000 mg PO TID x 7 days or Famciclovir 500mg PO TID VZIG should be offered to susceptible | Acyclovir 800 mg PO five times / day x 7 days | To be started within 24 hours of onset of rash.<br><br>Chickenpox during pregnancy does not justify termination without prior prenatal diagnosis as only. |
|------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|



|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
|                                                                                                                                    | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
|                                                                                                                                    | Date of Approval:30/04/2024             |
| <b>Hospital Infection Control</b>                                                                                                  | Date of Review:30/05/2025               |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Page <b>73</b> of <b>98</b>             |

|               |                                                                                                                                                                                                           |                                                                                                                                                                                             |                                                        |                                                                                                                                                                                                                      |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | Aciclovir<br>800mg Oral 5<br>times a day<br>IV<br>Acyclovir<br>recommen ded<br>for the<br>treatment of<br>severe<br>complicati<br>ons, > 24 hrs<br>from the onset<br>of rash,<br>antivirals not<br>useful | women < 10<br>days of the<br>exposure.<br>VZIG has no<br>role in<br>treatment<br>once the rash<br>appears.<br>The dose of<br>VZIG is 125<br>units / 10kg<br>not exceeding<br>625 units, IM. |                                                        | A minority of fetuses<br>infected develop fetal<br>varicella syndrome.                                                                                                                                               |
| Herpes zoster | Varicella zoster<br>virus                                                                                                                                                                                 | Valacyclovir<br>1000 mg PO TID<br>x 7 days                                                                                                                                                  | Acyclovir 800<br>mg PO five<br>times / day x 7<br>days | Begin within 72 hours<br>of onset of rash<br>Immunocompromised<br>patients: Acyclovir 10<br>mg/kg IV TID. Change<br>to Valacyclovir 1000<br>mg PO TID once<br>infection is controlled<br>Total duration 7-10<br>days |

#### BITES

|                                 |                                                                                                                                          |                                              |  |                                                                                 |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|--|---------------------------------------------------------------------------------|
| Bites (cat, dog,<br>human, rat) | Pasteurella<br>multocida,<br>Capnocytophag<br>a, Eikenella,<br>Strep viridans,<br>Spirillum<br>minus,<br>Streptobacillus<br>moniliformis | Amoxicillin-<br>Clavulanate 625<br>mg PO TID |  | Duration: 3 - 5 days<br>Administer Anti-rabies<br>vaccination as<br>appropriate |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|--|---------------------------------------------------------------------------------|

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue: 30/06/2024               |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation: 20/05/2024            |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval: 30/04/2024            |
|                                                                                                                                    | Date of Review: 30/05/2025              |
| Page 74 of 98                                                                                                                      |                                         |

## OTHER INFECTIONS

| Condition     | Most likely Microbial etiology | 1st Choice                                                                                                                                                                                                                                                                                                                                                                                          | Alternatives                                                                                                                                                                                                  | Comments                                                                                                                                                   |
|---------------|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Leptospirosis | Leptospira interrogans         | Mild - Doxycycline for 7 days<br><br>Mod or Severe - Penicillin G 15 lakh units IV Q6h                                                                                                                                                                                                                                                                                                              | Mild - Ampicillin, Amoxicillin<br><br>Mod or severe - Ceftriaxone 1gm IV OD or Ampicillin 0.5 to 1gm IV Q6h                                                                                                   | Monitor for Jarisch-Herxheimer reactions in patients treated with penicillin<br><br>Once daily ceftriaxone has been shown to be as effective as penicillin |
| Malaria       | P. vivax,<br><br>P. falciparum | <b>Vivax - Uncomplicated, not from chloroquine resistant region -</b><br>Chloroquine 600mg PO base (=1000mg salt), followed by 300mg base (=500mg salt) at 6, 24, 48<br><b>hrs. total dose = 1500mg base (=2500mg base), followed by Primaquine 30mg/day PO for 14 days</b><br><b>Vivax - Complicated, chloroquine resistance- Artesunate 2.4mg/kg at 0, 12, 24 hours and then OD for 5-7 days.</b> | Acute uncomplicated falciparum or oral step down for falciparum malaria - Artemether + Lumefantrine (40+480) combination (6 doses- 0,8,24,36,48 and 60 hrs)<br>Arterolane + Piperaquine once daily for 3 days | IV Quinine only used in severe malaria/ resistance, in ICU setting                                                                                         |

In pregnancy, antimalarial safety is not well defined. Mefloquine is Cat B, Chloroquine not found to be harmful, Quinine not contraindicated, Artemisinin compounds are accepted to use in second and third trimester. Doxycycline and Primaquine are contraindicated.

I<sup>st</sup> trimester - Quinine + clindamycin for 7 days, alternate therapy - Artesunate + Clindamycin.

II<sup>nd</sup> and III<sup>rd</sup> trimester - Artesunate + Clindamycin.

|                        |                                                                     |                                    |                                                                                  |                          |
|------------------------|---------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------|--------------------------|
| Rickettsial infections | Commonly - Scrub typhus- O. tsutsugamushi<br>Others – Spotted fever | Doxycycline 100mg PO BD for 7 days | Azithromycin 500mg PO OD for 3 days,<br>Chloramphenicol 500mg PO Q6H for 7 days. | Pregnancy - Azithromycin |
|------------------------|---------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------|--------------------------|

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
|                                                                                                                                    | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
|                                                                                                                                    | Date of Approval:30/04/2024             |
| <b>Hospital Infection Control</b>                                                                                                  | Date of Review:30/05/2025               |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Page <b>75</b> of <b>98</b>             |

|                        |                                                                   |                                                                        |                                                                       |                                                                                                                                                                   |
|------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Infective Endocarditis | group                                                             |                                                                        |                                                                       |                                                                                                                                                                   |
|                        | Viridians Streptococci (60-80%), Staphylococci, GNB, fungi, Mixed | Culture based, Vancomycin + Ceftriaxone                                | Step down based on cultures.                                          | 2 aerobic and 2 anaerobic bottles for culture at 30mins interval                                                                                                  |
|                        | Culture negative                                                  | Ampicillin - Sulbactam 3gm IV Q6h + Gentamycin 1mg/kg Q8hfor 4-6 weeks | Vancomycin + Gentamycin + Ciproflox <b>OR</b> Ceftriaxone + Ciproflox | Covers Enterococci, HACEK, NVS<br><br>Consider fungal, Brucella, other fastidious organisms & non-infective causes                                                |
| Febrile Neutropenia    | Non-high risk                                                     | Amoxiclav + Ciprofloxacin                                              | Cefixime + Ciprofloxacin                                              | Skin and CVC infection - Vancomycin/ Teicoplanin.<br><br>Consider empiric antifungals if persistently febrile and neutropenic after empiric antibacterial therapy |
|                        | High risk                                                         | Pip-Tazo or Cefoperazone Sulbactam                                     | Ceftazidime, Cefepime, Meropenem, Imipenem, Amikacin                  |                                                                                                                                                                   |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval:30/04/2024             |
|                                                                                                                                    | Date of Review:30/05/2025               |
| Page 76 of 98                                                                                                                      |                                         |

## 2.6 OBSTETRICS AND GYNAECOLOGICAL INFECTIONS:

| Condition        | Most likely Microbial etiology                                                                                                     | 1st Choice                                                                                                                                 | Alternatives         | Comments                                                                                                          |
|------------------|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------------------------------------------|
| Chorioamnionitis | Group B streptococcus, Gram negative bacilli, chlamydiae, ureaplasma and anaerobes, usually Polymicrobial                          | Clindamycin/ vancomycin/ teicoplanin and cefoperazone-sulbactam<br>If patient is not in sepsis then IV Ampicillin                          |                      | Preterm Birth, 9- 11% death rate in preterm infant's unfavorable neurologic outcome, lesser risk to term infants. |
| Septic abortion  | Bacteroides, Prevotella bivia,<br><br>Group B, Group A Streptococcus, Enterobacteriaceae, C. trachomatis, Clostridium perfringens. | Ampicillin 500 mg QID + Metronidazole 500mg IV TDS if patient has not taken any prior antibiotic (start antibiotic after sending cultures) | Ceftriaxone 2g IV OD |                                                                                                                   |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval:30/04/2024             |
|                                                                                                                                    | Date of Review:30/05/2025               |
| Page 77 of 98                                                                                                                      |                                         |

|                                                  |                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                |  |  |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--|--|
| Endomyometritis and Septic Pelvic Vein Phlebitis | Bacteroides,<br>Prevotella bivia,<br><br>Group B, Group A<br>Streptococcus,<br>Enterobacteriaceae,<br><i>C. trachomatis</i> ,<br><i>Clostridium perfringens</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Same as above. |  |  |
| <b>Obstetric Sepsis during pregnancy</b>         | Group A beta-haemolytic<br>Streptococcus,<br><i>E.coli</i> , anaerobes.                                                                                         | If patient is in shock and blood culture reports are pending, then start Piperacillin-Tazobactam<br><br>OR<br><br>per the report. If patient has only fever, with no features of severe sepsis start Amoxicillin Clavulanate oral 625TDS/IV 1.2 gm TDS<br>Or Ceftriaxone 2gm IV OD+ Metronidazole 500mg IV TDS<br><br>+/-gentamicin<br><br>7mg/kg/day OD if admission needed. MRSA cover may be required if suspected or colonized (Vancomycin/ Teicoplanin) |                |  |  |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval:30/04/2024             |
|                                                                                                                                    | Date of Review:30/05/2025               |
| Page <b>78</b> of <b>98</b>                                                                                                        |                                         |

|                                             |                                                                                                                           |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                            |                                                          |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Obstetric Sepsis following pregnancy</b> | S. pyogenes, E. coli, S. Aureus, S.pneumoniae, Methicillin-resistant S. aureus (MRSA), C. septicum & Morganella morganii. | Same as above                                                                                                                                                                                                                                                                        | Genital tract<br><br>Mastitis,<br><br>UTI<br><br>Pneumonia<br>Skin<br><br>and<br><br>soft<br><br>tissue<br><br>(IV<br>site,<br>surgical<br>site,<br>drain<br>site<br>etc.) |                                                          |
| <b>Tuberculosis in pregnancy</b>            | Like NON - PREGNANT population with some exceptions                                                                       | recommended by US FDA)<br>Mother and baby should stay together, and the baby should continue to breastfeed. Pyridoxine supplementation is recommended for all pregnant or breastfeeding women taking isoniazid as well as to neonate who are being breast fed by mothers taking INH. |                                                                                                                                                                            | Very small chance of transmission of infection to fetus. |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval:30/04/2024             |
|                                                                                                                                    | Date of Review:30/05/2025               |

Page 79 of 98

|                                                      |                 |                                                                                                                                                                                                                                                                                                                                     |  |                                                                                                                                                                                                     |
|------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Acute Toxoplasmosis in pregnancy</b>              |                 | every two weeks If PCR Positive.<br>>18 weeks gestation and documented fetal infection by positive amniotic fluid PCR.<br>Pyrimethamine 50 mg Oral BD x 2 days then 50 mg OD PLUS Sulphadiazine 75 mg/kg Oral x 1 dose then 50mg/kg bd PLUS Folinic Acid (10-20 mg Oral daily) for minimum of 4 weeks or for duration of pregnancy. |  |                                                                                                                                                                                                     |
| <b>Candidiasis</b>                                   | Candida species | Fluconazole oral 150 mg single dose<br>For milder cases- Intravaginal agents as creams or suppositories Clotrimazole, Miconazole, Nystatin. Intravaginal azoles, single dose to 7-14 days.                                                                                                                                          |  | Non-pregnant- If recurrent candidiasis, (4 or more episodes/year) 6 months suppressive treatment with Fluconazole 150 mg oral once a week or Clotrimazole vaginal suppositories 500 mg once a week. |
| <b>Cervicitis/Urethritis Mucopurulent gonococcal</b> | Polymicrobial   | Ceftriaxone 250 mg IM<br>Single dose + Azithromycin 1 gm single dose OR Doxycycline 100mg BD x 7 day                                                                                                                                                                                                                                |  |                                                                                                                                                                                                     |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval:30/04/2024             |
|                                                                                                                                    | Date of Review:30/05/2025               |

Page 80 of 98

|                                                                             |                                                      |                                                                                                                                                                                                                                    |  |                                                                                        |
|-----------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------|
| <b>Pelvic Inflammatory Disease (Salpingitis &amp; tubo-ovarian abscess)</b> | S. aureus, Enterobacteriaceae, gonococci, gardenella | OP: Ceftriaxone 250 mg IM/IV single dose plus +/- Metronidazole 500 mg BD x 14 days<br>Plus Doxycycline 100 mg BD x 14 Days<br>IP: Clindamycin + ceftriaxone till patient admitted then change to OPD treatment                    |  | Drainage of tubo- ovarian abscess wherever indicated<br>Evaluate and treat sex partner |
| <b>Mastitis without abscess</b>                                             | S. aureus                                            | Amoxycillin clavulanate/Cephalexin 500 mg QID/BD OR Ceftriaxone 2 gm OD OR MRSA- based on sensitivities Add Clindamycin 300 QID OR Vancomycin 1 gm IV 12 H /Teicoplanin 12mg/kg IV 12 hourly x 3 doses followed by 6 once daily IV |  |                                                                                        |
| <b>Mastitis with abscess</b>                                                |                                                      | Drainage with antibiotic cover for MRSA Clindamycin 300 QID OR Vancomycin 15mg/kg IV 12 hourly (max 1gm 12H)/ Teicoplanin 12mg/kg IV 12 hourly x 3 doses followed by 6 mg once daily IV                                            |  |                                                                                        |

## 2.6 Paediatric infections Dosages

| Sl No. | Drug         | Route | <2 Kg                |                     | >2kg                |                     |
|--------|--------------|-------|----------------------|---------------------|---------------------|---------------------|
|        |              |       | 0-7 Days             | >7 Days             | 0-7 Days            | >7 Days             |
| 1      | Ampicillin   | IV    | 25mg/Kg/D ose (q12h) | 25mg/Kg/ Dose (q8h) | 50mg/Kg/ Dose(q12h) | 35mg/Kg/D ose (q8h) |
| 2      | Netilmycin   | IV    | 5 mg/Kg/Dose (OD)    | 5 Mg/Kg/Do se (OD)  | 5 Mg/Kg/Do se (OD)  | 5 Mg/Kg/Dos e (OD)  |
| 3      | Piperacillin | IV    | 50mg/Kg/D ose        | 50mg/Kg/            | 50mg/Kg/ Dose       | 50mg/Kg/D ose       |



|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
|                                                                                                                                    | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
|                                                                                                                                    | Date of Approval:30/04/2024             |
| <b>Hospital Infection Control</b>                                                                                                  | Date of Review:30/05/2025               |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Page <b>81</b> of <b>98</b>             |

|   |                   |    |                  |                |                  |                |
|---|-------------------|----|------------------|----------------|------------------|----------------|
|   | Tazobactam        |    | (BD)             | Dose (TID)     | (BD)             | (TID)          |
| 4 | Amikacin          | IV | 7.5 Mg/Kg (q12h) | 7.5mg/Kg (q8h) | 7.5 Mg/Kg (q12h) | 7.5mg/Kg (q8h) |
| 5 | Meropenem         | IV | 20mg/Kg (q12h)   | 20mg/Kg (q8h)  | 20mg/Kg (q12h)   | 20mg/Kg (q8h)  |
| 6 | Vancomycin        | IV | 15 mg/Kg (q12)   | 15 mg/Kg (q8h) | 15 mg/Kg (q12h)  | 15 mg/Kg (q8h) |
| 7 | Cefotaxime Sodium | IV | 50mg/Kg (q12h)   | 50mg/Kg (q8h)  | 50mg/Kg (q12h)   | 50mg/Kg (q8h)  |
| 8 | For Meningitis:   |    | 50mg/Kg (q6h)    | 50mg/Kg (q6h)  | 50mg/Kg (q6h)    | 50mg/Kg (q6h)  |
|   | Cefotaxime Sodium | IV |                  |                |                  |                |

|                                                   |                                                     |                   |
|---------------------------------------------------|-----------------------------------------------------|-------------------|
|                                                   | <b>Sepsis +Pneumonia</b>                            | <b>Meningitis</b> |
| First Line (Early Onset Sepsis/Late Onset Sepsis) | Netilmycin ± Ampicillin                             | Cefotaxime Sodium |
| Second Line<br><br>(Hospital Acquired)            | Piperacillin<br><br>Tazobactam/Vancomycin +Amikacin | Cefotaxime Sodium |
| Third Line                                        | Meropenem ± Vancomycin                              | -                 |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue: 30/06/2024               |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation: 20/05/2024            |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval: 30/04/2024            |
|                                                                                                                                    | Date of Review: 30/05/2025              |
| Page <b>82</b> of <b>98</b>                                                                                                        |                                         |

## 2.6 PAEDIATRIC INFECTIONS:

| Condition                                                                               | Most likely microbial etiology              | 1 <sup>st</sup> choice                                                                                                                                                                                                                                                    | Alternative                                                                                                                                                                                                                    | Comments                      |
|-----------------------------------------------------------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Bacterial Meningitis                                                                    | Gram- negative bacilli or Staphylococcus sp | For Infant below 2 months age (more than 2kg):<br>Inj. Cefotaxime 50 mg/kg/dose (if <7 days - 12hrly, >7days - 8hrly)<br>+ Inj. Gentamicin 5 mg/kg/dose (if <7 days - 24hrly, >7days - 24hrly)<br>For 2 months and above: Inj Ceftriaxone (100mg/kg/day-2 divided dosage) | Meropenem, 20 mg/kg/dose (if <7 days - 12hrly, >7days - 8hrly)<br><br>+ Vancomycin, 15 mg/kg/dose (if <7 days - 12hrly, >7days - 8hrly)<br>Meropenem (120 mg/kg/day in 3 div doses) + Vancomycin (60mg /kg/day in 4 div doses) | Duration of therapy 2-3 weeks |
| <b>Lower Respiratory Tract infection:</b><br><br>a) <b>Community acquired pneumonia</b> |                                             | For Severe Pneumonia (Children with respiratory distress requiring IP care) -<br>Under 2 months of age:<br><br>Inj. Cefotaxime and Gentamicin for 10 days                                                                                                                 |                                                                                                                                                                                                                                |                               |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
|                                                                                                                                    | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
|                                                                                                                                    | Date of Approval:30/04/2024             |
| <b>Hospital Infection Control</b>                                                                                                  | Date of Review:30/05/2025               |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Page <b>83</b> of <b>98</b>             |

|                       |  |                                                                                                                                                                                                                                                                                                                                                                                                                              |            |  |
|-----------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--|
| <b>b/HA pneumonia</b> |  | <p>Over 2 months of age: Inj. Ampicillin (50mg/kg/dose 6h) + Gentamicin (7.5mg/kg/day OD IM or IV) is used.</p> <p>Inj. Ampicillin can be switched to Oral Amoxicillin (45mg/kg/day TDS) once child is stable and able to take oral feeds</p> <p>In case the patient has severe sepsis/ septic shock, Inj. Piperacillin + Tazobactam (90mg/kg/dose 6h) + MRSA cover with IV Vancomycin (15mg/kg/dose 6h) is recommended.</p> |            |  |
| <b>c) Empyema</b>     |  | <p>Anti-Staphylococcal Penicillin (Cloxacillin 100-200 mg/kg/day)</p> <p>+</p> <p>Ceftriaxone/Co-Amoxyclav</p>                                                                                                                                                                                                                                                                                                               | Vancomycin |  |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue: 30/06/2024               |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation: 20/05/2024            |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval: 30/04/2024            |
|                                                                                                                                    | Date of Review: 30/05/2025              |

Page 84 of 98

|                                            |                                                                                       |                                                                                                                                                                                                      |                                                                                                                                                                                                                                          |  |
|--------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>d) Ventilator Associated Pneumonia:</b> | Pseudomonas aeruginosa<br>OR<br>Klebsiella pneumoniae (ESBL) Or Acinetobacter species | Beta Lactam + Beta lactamase inhibitor (Piperacillin – Tazobactam)<br>Plus<br>Either<br>Aminoglycoside (Amikacin, Gentamicin, or Tobramycin) OR Antipseudomonal Fluroquinolone (Cipro/ Levofloxacin) | Meropenem – 60 mg/kg/day I/V every 8 hrly AND<br>Vancomycin - 40 mg/kg/day I/V every 6 - 8 hrly.<br>OR<br>Colistin base IV., 2.5 – 5 mg/kg/day I/V every 6 – 12 hrly (1mg= 30000 IU) AND Vancomycin - 40 mg/kg/day I/V every 6 - 8 hrly. |  |
|                                            | Methicillin – resistant Staphylococcus aureus (MRSA)                                  | Vancomycin or Linezolid                                                                                                                                                                              |                                                                                                                                                                                                                                          |  |

|                                           |                       |                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                             |  |
|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>UPPER RESPIRATORY TRACT INFECTIONS</b> | Group A Streptococcus | Penicillin V (Oral) for (<30kg) 250 mg BID x 10 days<br>(>30kg) 500 mg BID x 10 days<br>OR<br>Amoxicillin (Oral) for (<30kg) 40 mg/kg/day x 10 days, (>30kg) 250 mg TID, can be given BID | If the patient is Penicillin allergic,<br>Erythromycin (oral) 40-50 mg/kg/day BID or TID x 10 days<br>OR<br>Azithromycin (oral) 12 mg/kg OD x 5 days<br>OR<br>First Generation Cephalosporin (oral) (10 days) (if only, it is non-Type 1 allergy to penicillin)<br>Cefaclor (20-40 mg/kg/d in 3 divided doses) / Cephalexin (50 mg/kg/d in 3 divided doses) |  |
|                                           |                       |                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                             |  |

**a) Bacterial Pharyngotonsillitis**

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
|                                                                                                                                    | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
|                                                                                                                                    | Date of Approval:30/04/2024             |
| <b>Hospital Infection Control</b>                                                                                                  | Date of Review:30/05/2025               |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Page <b>85</b> of <b>98</b>             |

|                            |  |                                                                                                                                                                                                                                                                          |                                                                                 |                                                        |
|----------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------|
| b) Faucial diphtheria      |  | Inj. Crystalline Penicillin (100,000-150,000 Units/kg/day I.V 6h) for 10 days.<br>In case of penicillin allergy, Erythromycin (40-50mg/kg/day TDS/QID for 14 days or Azithromycin 10mg/kg/day OD for 5 days                                                              | Inj. Procaine Penicillin 25,000-50,000 U/kg/day BD maximum of 12 lakh units IM. |                                                        |
| c) Acute otitismedia (AOM) |  | Oral Amoxicillin (45mg/kg/day TDS/50-60mg/kg/day in two divided doses) for 7 to 10 days<br>For severe cases (Severe Otagia and/ or Temp >390C), Co-Amoxicillin Clavulanate (45 mg/kg/day PO BD)<br>In case of penicillin allergy, Cefidininir 14mg/kg in 2 divided doses | Inj. Ceftriaxone 75 mg/kg/day OD                                                | Duration of treatment is recommended for 7 to 10 days. |

|                                                                                                                                    |  |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL |  | Doc No: KASPP/AB/01                     |
|                                                                                                                                    |  | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                  |  | Revision Date and Status: 30/05/2024/03 |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           |  | Date of Creation:20/05/2024             |
|                                                                                                                                    |  | Date of Approval:30/04/2024             |
|                                                                                                                                    |  | Date of Review:30/05/2025               |
| Page <b>86</b> of <b>98</b>                                                                                                        |  |                                         |

|                                                                                                       |  |                                                                                    |                                                 |  |
|-------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------|-------------------------------------------------|--|
| <b>d)</b><br><b>Mastoiditis</b><br><b>and</b><br><b>other Acute</b><br><b>Ear</b><br><b>Infection</b> |  | Inj. Amoxicillin-clavulanate                                                       | 3rd Gen Cephalosporin (Ceftriaxone/cefotaxime). |  |
| <b>e) Acute Sinusitis with URI</b>                                                                    |  | Oral Amoxicillin (45 mg/kg/day TDS) for 7-10 days                                  | Inj. Ceftriaxone 75 mg/kg/day OD                |  |
|                                                                                                       |  | For severe cases, Amoxicillin Clavulanate (45 mg/kg/day oral BD)                   |                                                 |  |
| <b>f) Ludwig's Angina</b>                                                                             |  | Clindamycin IV 8 hourly OR Amoxicillin-Clavulanate IV                              | Piperacillin-Tazobactam IV 6 hourly             |  |
| <b>g) Pertussis</b>                                                                                   |  | Erythromycin for 14 days, OR Azithromycin for 5 days, OR Clarithromycin for 7 days |                                                 |  |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval:30/04/2024             |
|                                                                                                                                    | Date of Review:30/05/2025               |
| Page 87 of 98                                                                                                                      |                                         |

## 2.7 GASTROINTESTINAL DISEASES

|                         |  |                                                                                                                                                                                                                                                                          |                                                                                                                                                                                     |  |
|-------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>a) Dysentery</b>     |  | For IP - Inj<br>Ceftriaxone<br>(100mg/kg) for<br>5-7 days.<br>For OP - Oral<br>Cefixime<br>(8mg/kg/day<br>BD) OR<br>Azithromycin 10-<br>20 mg/ kg (ceiling<br>dose of 1 gm) for<br>5 days.                                                                               | Azithromycin 10-20<br>mg/ kg (ceiling dose<br>of 1 gm) for 5 days.                                                                                                                  |  |
| <b>b) Cholera</b>       |  | Single dose oral<br>azithromycin 10<br>mg/ kg (ceiling<br>dose of 1 gm).                                                                                                                                                                                                 | Doxycycline (50mg<br>for less than 3 years<br>and 100 mg for those<br>above).                                                                                                       |  |
| <b>c) Enteric Fever</b> |  | For OP - Oral<br>Cefixime 20<br>mg/kg/day (max<br>dose of 1200) for<br>14 days or<br>Azithromycin<br>10-20 mg/kg<br>(ceiling dose of 1<br>gm) for 7-10<br>days.<br>For IP –<br>Inj Ceftriaxone<br>100 mg/ kg/day<br>and shift to oral<br>cefixime once<br>fever resolves | Ofloxacin 15mg/kg/d<br>in two divided doses<br>for 10-14 days.<br>OR<br><br>Chloramphenicol (50-<br>75mg/kg/day PO) x 14<br>days c) TMP-SMX (8<br>mg/kg/day of TMP<br>PO) x 14 days |  |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval:30/04/2024             |
|                                                                                                                                    | Date of Review:30/05/2025               |
| Page <b>88</b> of <b>98</b>                                                                                                        |                                         |

|                         |  |                                                                                                                                                                                                                                |                                |                                                                                                                                                                                                        |
|-------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>d) Peritonitis</b>   |  | Inj. Ampicillin +<br>Aminoglycoside                                                                                                                                                                                            | Inj. cefotaxime                |                                                                                                                                                                                                        |
| <b>e) Liver Abscess</b> |  | Amoxycillin+Cl<br>avulanate /<br>3rd generation<br>cephalosporin<br><br>+<br><br>Aminoglycoside<br><br>Metronidazole<br>(30-50 mg/kg/d<br>in 3 divided<br>doses for 10-14<br>days) is added if<br>Amoebic abscess<br>suspected | Piperacillin-<br>Tazobactam IV | Ultrasound<br>guided<br>drainage<br>indicated in<br>large<br>abscesses,<br>signs of<br>imminent<br>rupture and<br>noresponse<br>to medical<br>treatment.<br><br>Duration of<br>therapy is 4-<br>6weeks |

|                                                                                                |  |                                                                                              |  |                                         |
|------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------|--|-----------------------------------------|
| <b>URINARY<br/>TRACT<br/>INFECTION<br/>(UTI)<br/><br/>a) Uncomplicated<br/>UTI (age&gt; 2)</b> |  | Oral<br>Cotrimoxazole<br>(8-10mg of<br>TMP<br>component)<br>/kg/day oral BD<br><br><b>OR</b> |  | Duration of<br>therapy is 7-<br>10 days |
|------------------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------|--|-----------------------------------------|



|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval:30/04/2024             |
|                                                                                                                                    | Date of Review:30/05/2025               |
| Page <b>89</b> of <b>98</b>                                                                                                        |                                         |

|                                                                                                     |  |                                                                                                                                      |  |                                   |
|-----------------------------------------------------------------------------------------------------|--|--------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------|
| months with lower UTI, without any urinary tract obstruction)                                       |  | Cefixime (8-10 mg/kg/day BD) to be given for 7-10 days<br><b>OR</b><br>Co-Amoxycillin + Clavulanic Acid (30-50 mg of Amoxicillin)    |  |                                   |
| <b>b) Complicated / Severe UTI</b>                                                                  |  | Inj.<br>Cefotaxime (150-200mg/kg/day) 8h<br><b>OR</b> Inj.<br>Ceftriaxone (100mg/kg/day) OD<br><b>OR</b> Inj.<br>Amikacin 15mg/kg OD |  | Duration of therapy is 10-14 days |
| <b>c) In Immunocompromised host / severe systemic sepsis or as second line for complicated UTI-</b> |  | Inj.<br>Piperacillin Tazobactam 90mg/kg/dose IV 6h) or Inj.<br>Meropenem (20-40mg/kg/dose 8h)                                        |  | Duration of therapy is 10-14 days |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue: 30/06/2024               |
|                                                                                                                                    | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation: 20/05/2024            |
|                                                                                                                                    | Date of Approval: 30/04/2024            |
| <b>Hospital Infection Control</b>                                                                                                  | Date of Review: 30/05/2025              |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Page <b>90</b> of <b>98</b>             |

|                            |  |                                                                                                   |                                                                                                                                                                                                                                                                                       |                                     |
|----------------------------|--|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>FEBRILE NEUTROPENIA</b> |  | Ceftazidime (150 mg/kg/day in 3 div doses) +<br><br>Amikacin (15-20mg/kg/day in 2 or 3 div doses) | Piperacillin + Tazobactam (200-300 mg/kg/day IV in 3-4 div doses) + Vancomycin (40 mg/kg/day IV in 4 divided doses)<br>OR<br>Meropenem (60 mg/kg/day in 3 div doses) PLUS Amphotericin B (1 mg/kg/day IV for 2 weeks) or liposomal Amphotericin B 1-5 mg/kg/day, usually 3 mg/kg/day. | Duration of therapy is 7 days       |
| <b>BRAIN ABSCESS</b>       |  | Inj. Ceftriaxone + Vancomycin                                                                     | Inj. Teicoplanin + Metronidazole                                                                                                                                                                                                                                                      | Duration of therapy is 3 to 4 weeks |
| <b>OSTEOMYELITIS</b>       |  | Co-Amoxy clavulanic + Gentamicin                                                                  |                                                                                                                                                                                                                                                                                       |                                     |
| <b>SEPTIC ARTHRITIS</b>    |  | Inj. Co-Amoxy clavulanic + Gentamicin                                                             | Inj. Ceftriaxone//cefotaxime +/- Vancomycin                                                                                                                                                                                                                                           |                                     |
| <b>TETANUS</b>             |  | Inj. Crystalline Penicillin (2 lac IU /kg/ d /12 hourly)                                          | Inj. Metronidazole (mg/kg/d)                                                                                                                                                                                                                                                          |                                     |
| <b>ACUTE ENDOCARDITIS</b>  |  | Inj. Crystalline Penicillin/ampicillin + Gentamicin                                               |                                                                                                                                                                                                                                                                                       | Duration of therapy is 3-4 weeks    |

|                                    |  |                                                                                                                           |                                                                                                                                              |  |
|------------------------------------|--|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>ACUTE RHEUMATIC FEVER (ARF)</b> |  | Benzathine Penicillin G<br>weight $\geq$ 27 kg: - 1.2 million units<br>weight < 27 kg: - 0.6 million units<br>SINGLE DOSE | Azithromycin 12.5 mg/kg/day divided OD- 5 days<br>OR<br>Amoxicillin 25–50mg/kg/day divided<br>Adult dose 750-1500 mg/day TDS - 10 days<br>OR |  |
|------------------------------------|--|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|--|

|                                                                                                                                                                                                                                                                                                                             |  |  |  |                                          |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|------------------------------------------|--|--|
| <div><b>KIMS ALSHIFA</b><br/>SUPER SPECIALITY HOSPITAL</div> <div></div> <div><b>Hospital Infection Control</b></div> <div><b>ANTIBIOTIC POLICY</b></div> |  |  |  | Doc No: KASPP/AB/01                      |  |  |
|                                                                                                                                                                                                                                                                                                                             |  |  |  | Date of Issue:30/06/2024                 |  |  |
|                                                                                                                                                                                                                                                                                                                             |  |  |  | Revision Date and Status: 30/05/2024/03  |  |  |
|                                                                                                                                                                                                                                                                                                                             |  |  |  | Date of Creation:20/05/2024              |  |  |
|                                                                                                                                                                                                                                                                                                                             |  |  |  | Date of Approval:30/04/2024              |  |  |
|                                                                                                                                                                                                                                                                                                                             |  |  |  | Date of Review:30/05/2025                |  |  |
|                                                                                                                                                                                                                                                                                                                             |  |  |  | Page <b>91</b> of <b>98</b>              |  |  |
|                                                                                                                                                                                                                                                                                                                             |  |  |  | Cephalexin 15-20 mg/kg/dose BD - 10 days |  |  |

## 2.6 ANTIMICROBIALS DOSING IN RENAL INSUFFICIENCY

Cockcroft-Gault equation to calculate creatinine clearance (CrCl) for drug dosing in renal impairment:  $\text{CrCl (ml/min)} = (140 - \text{age}) \times \text{weight (kg)} / 72 \times \text{serum creatinine (mg/dl)}$  (x 0.85 if female)

| Anti-microbial agent                                                                                  | CrCl (ml/min)                                               |               |                                                  |                                                    | Comments                     |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|---------------|--------------------------------------------------|----------------------------------------------------|------------------------------|
|                                                                                                       | Normal dose                                                 | Mild (20-50)  | Moderate (10-20)                                 | Severe (<10)                                       |                              |
| Acyclovir IV                                                                                          | 10 mg/kg q8h                                                | 10 mg/kg q12h | 10mg/kg q24h                                     | 5mg/kg q24h                                        | AD                           |
| Acyclovir PO                                                                                          | Simplex: 200-400 mg q6h<br><br>Zoster: 800 mg 5 times a day | Normal dose   | Simplex: 200- 400mg q6h<br><br>Zoster: 800mg q8h | Simplex: 200- 400mg q12h<br><br>Zoster: 800mg q12h | AD                           |
| Amikacin IV                                                                                           | 15 mg/kg q24h                                               | 7.5mg/kg q24h | 7.5mg/kg q24- 48h                                | 7.5mg/kg q48- 72h                                  | Supplement HD:7.5mg/kg AD    |
| Amoxicillin PO                                                                                        | 500 mg - 1g q8h                                             | Normal        | 500mg - 1g q12h                                  | 500mg - 1g q12h                                    | AD                           |
| Amphoteric in IV                                                                                      | 1 mg/kg q24h                                                | Normal        | Consider lipid formulations                      | Consider lipid formulations                        | No supplementati on after HD |
| <b>Amphotericin is highly NEPHROTOXIC. Consider using Liposomal Amphotericin. Monitor CrCl daily.</b> |                                                             |               |                                                  |                                                    |                              |
| <b>Liposomal Amphotericin IV</b>                                                                      | 3–5 mg/kg q24h                                              | Normal        | Normal                                           | Normal                                             |                              |

|                                                                                                                                   |                                         |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                   | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                 | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                   | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                          | Date of Approval:30/04/2024             |
|                                                                                                                                   | Date of Review:30/05/2025               |
|                                                                                                                                   | Page <b>92</b> of <b>98</b>             |

|                                                                        |                                                                        |        |                                                                  |                                                                           |             |
|------------------------------------------------------------------------|------------------------------------------------------------------------|--------|------------------------------------------------------------------|---------------------------------------------------------------------------|-------------|
| <b>Ampicillin IV</b>                                                   | 1-2g Q4-6h                                                             | Normal | Normal                                                           | Normal                                                                    | AD          |
| <b>Anidulafungin IV</b>                                                | 200 mg for 1 day then 100 mg q24h                                      | Normal | Normal                                                           | Normal                                                                    |             |
| <b>Artemether with</b>                                                 | 20mg/120mg tabs: 4 tabs q12h for 3 days                                | Normal | Normal                                                           | Normal                                                                    | <10 Monitor |
| <b>Artesunate IV</b>                                                   | 2.4 mg/kg at 0, 12 and 24 h then q24h                                  | Normal | Normal                                                           | Normal                                                                    |             |
| <b>Azithromycin IV + PO</b>                                            | IV: 500 mg q24h<br>PO:500mg-1g Q24h                                    | Normal | Normal                                                           | Normal                                                                    |             |
| <b>Benzyl penicillin (CP) IV</b>                                       | 20-40L units q4h                                                       | Normal | 10-20L q4h                                                       | 5 - 10 L q4h                                                              | AD          |
| <b>Caspofungin IV</b>                                                  | 70 mg for 1 day then 50 mg q24h                                        | Normal | Normal                                                           | Normal                                                                    |             |
|                                                                        | [If >80 kg 70 mg q24h]                                                 |        |                                                                  |                                                                           |             |
| <b>Cefazolin IV</b>                                                    | 1-2g q8h                                                               | Normal | 1-2g q12h                                                        | 1g q24h                                                                   | AD          |
| <b>Cefoperazon e + Sulbactam IV (1.5g = 1g + 500 mg, 3g = 2g + 1g)</b> | 3g q12h but severe / less sensitive/neutr openia 3g q8h to max 6g q12h | Normal | 3g q12h plus if severe Extra Cefoperazone 2g q24h to max 2g q12h | 1.5g q12hplus extra cefoperazone1g q12h; if severe 2g q12h to max 3g q12h | AD          |
| <b>Ceftriaxone IV</b>                                                  | 1-2g q12-24h                                                           | Normal | Normal                                                           | Normal                                                                    |             |
| <b>Ciprofloxacin IV + PO</b>                                           | IV 400 mg q8h<br><br>PO 750 mg q12h                                    | Normal | IV 200-400 mg q12h<br><br>PO 250-500 mg q12h                     | IV 200-400 mg q12h<br><br>PO 250-500 mg q12h                              |             |
| <b>Clarithromycin IV + PO</b>                                          | IV 500 mg q12h<br>PO 500 mg q12h                                       | Normal | 250-500 mg q12h                                                  | 250-500mg q24h                                                            | AD          |
| <b>Clindamycin IV + PO</b>                                             | IV 600mg q6-8h po 300-450 MG Q6H                                       | Normal | Normal                                                           | Normal                                                                    |             |
| <b>Cloxacillin IV + PO</b>                                             | IV 1-2g q4-6h<br><br>PO 500 mg-1g q6h                                  | Normal |                                                                  | Normal (Max 6g/day)                                                       |             |

|                                                                                                                                    |  |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL |  | Doc No: KASPP/AB/01                     |
|                                                                                                                                    |  | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                  |  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    |  | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           |  | Date of Approval:30/04/2024             |
|                                                                                                                                    |  | Date of Review:30/05/2025               |
|                                                                                                                                    |  | Page <b>93</b> of <b>98</b>             |

|                                                                                               |                                                                                                                                                                                                                                                                                        |                                                                                                                                                                           |                                                                                                  |                                                                         |                                                                                   |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Co-Amoxiclav IV + PO</b>                                                                   | IV-1.2gm q8h<br>PO 1000 mg q12h                                                                                                                                                                                                                                                        | Normal                                                                                                                                                                    | Normal                                                                                           | IV 1.2 g q24h<br>PO 625 mg q8-12h                                       | AD                                                                                |
| <b>Colistimethate sodium IV</b>                                                               | 9MU Loading dose (LD)<br>[If <60kg 0.15MU/kg]<br>THEN after 12 hours                                                                                                                                                                                                                   | CrCl-60-80: Normal<br><br>CrCl-50-59: Normal LD; Then, 4MU q12h [If <60kg 0.044MU/kg q8h]<br>CrCl-20- 49: Normal LD; Then, 2MU q8h. (If >60kg) [If <60kg 0.038MU/kg q8h]. | IV 1.2g q12h<br><br>PO 625 mg q8- 12h<br><br>Normal LD then 2 MU q12h. [If <60kg .038MU/kg q12h] | Normal LD then 1.5 MU q12h [If <60kg 0.023MU/kg q12h]                   | 2MU q12h on dialysis days. [If <60kg 0.038MU/kg q12h]<br><br>NB If <60kg round to |
| <b>Colistimethate sodium NEB</b>                                                              | 1-2MU q12h                                                                                                                                                                                                                                                                             | 1 MU q12h                                                                                                                                                                 |                                                                                                  | 1 MU q24h                                                               | AD                                                                                |
| <b>Co-trimoxazole IV + PO (DS = 960 mg = 800mg sulfamethoxazole and 160 mg trimethoprim )</b> | PCP: (TMP 15mg/kg/day) or 1 DS (960mg) tab per 10 kg/day in 2-4 divided doses.<br>Meliodosis: 1920mg (2 DS, 960mg tabs) q12h<br>Nocardiosis: (TMP 10-15 mg/kg/day) in 2-4 divided doses<br>Toxoplasmosis : (TMP 5mg/kg)q12h<br>Prophylaxis: 960mg q24h or 960mg q12- 24h 3 days a week | CrCl-30-50: Normal                                                                                                                                                        | 1 MU q12h<br>CrCl-15-30<br>PCP: Normal for first 3days followed by 50% of usual dose             | CrCl<15<br>All infections: 50%of usual doses<br><br>Prophylaxis: Normal | AD                                                                                |
| <b>Dicloxacillin IV + PO</b>                                                                  | IV 1-2g q4-6h<br>PO 500 mg-1g q6h                                                                                                                                                                                                                                                      | Normal                                                                                                                                                                    | Normal                                                                                           | Normal (Max 6g/day)                                                     |                                                                                   |
| <b>Doxycycline IV + PO</b>                                                                    | 100 mg q12h                                                                                                                                                                                                                                                                            | Normal                                                                                                                                                                    | Normal                                                                                           | Normal                                                                  |                                                                                   |

|                                                                                                                                   |                                         |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                   | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                 | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                   | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                          | Date of Approval:30/04/2024             |
|                                                                                                                                   | Date of Review:30/05/2025               |
|                                                                                                                                   | Page <b>94</b> of <b>98</b>             |

|                                   |                                                                   |                                                                                                                 |                                                   |                                                                                                          |                                                |
|-----------------------------------|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <b>Ertapenem IV</b>               | 1g q24h                                                           | Normal                                                                                                          | Normal                                            | 500 mg q24h                                                                                              | AD                                             |
| <b>Ethambutol PO</b>              | 20 mg/kg q24h                                                     | Normal                                                                                                          | 20mg/kg q24-36h                                   | 20mg/kg q48h                                                                                             | AD                                             |
| <b>Fluconazole IV + PO</b>        | 50-800 mg daily                                                   | Start normal dose then 50%                                                                                      | Start normal dose then 50%                        | Start normal dose then 50%<br>HD: 100% after each dialysis [on non-dialysis days dose according to CrCl) | AD.<br>Single dose: Normal<br><br>CRRT: Normal |
| <b>Ganciclovir IV (Induction)</b> | 5 mg/kg q12h                                                      | CrCl-50-69<br>2.5 mg/kg q12h<br>CrCl-25-49 2.5 mg/kg q24h                                                       | CrCl-10-24<br>1.25 mg/kg q24h                     | 1.26 mg/kg 3 times a week                                                                                | AD                                             |
| <b>Gentamicin IV</b>              | Once daily: 5-7 mg/kg q24h<br>Infective endocarditis: 3mg/kg q24h | ‘Once Daily’:<br>60-79<br>4mg/kg q24h,<br>40-59<br>3.5mg/kg q24h,<br>30-39<br>2.5mg/kg q24h;<br>IE: 2mg/kg q24h | ‘Once daily’:<br>3mg/kg q48h<br>IE: 1mg/kg q24h   | ‘Once daily’:<br>2mg/kg q48-72h<br>IE: 1mg/kg q48h                                                       | AD                                             |
| <b>Imipenem IV</b>                | 500 mg–1g q6h                                                     | CrCl 41-70<br>500-750 mg q8h<br>CrCl 21-40<br>250-500 mg q6h                                                    | 250-500 mg q12h                                   | Risk of seizures – use Meropenem                                                                         | AD                                             |
| <b>Isoniazid PO</b>               | 5 mg/kg q24h                                                      | Normal                                                                                                          | Normal                                            | 150-300 mg q24h                                                                                          | AD                                             |
| <b>Itraconazole PO</b>            | 200 mg q12h                                                       | Normal                                                                                                          | Normal                                            | 100mg q12h                                                                                               | AD                                             |
| <b>Valganciclovir IV</b>          | Induction 900 mg q12h<br>Maintenance 900 mg q24h                  | CrCl-40-59<br>I-450 mg q12h<br>M 450 mg 24h<br>25-39<br>I 450 mg q24h<br>M 450 mg q48h                          | CrCl-10-24 I 450 mg q48h<br>M 450 mg twice Weekly | Not recommended<br>Limited data:450 mg 2-3 times a week, 450 mg 1-2 times a week                         | AD                                             |

|                      |                         |              |              |                          |                |
|----------------------|-------------------------|--------------|--------------|--------------------------|----------------|
| <b>Vancomycin IV</b> | 20-30mg/kg loading (max | 15mg/kg q24h | 15mg/kg q48h | 15mg/kg once followed by | Monitor trough |
|----------------------|-------------------------|--------------|--------------|--------------------------|----------------|

|                                                                                                                                    |  |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL |  | Doc No: KASPP/AB/01                     |
|                                                                                                                                    |  | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                  |  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    |  | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           |  | Date of Approval:30/04/2024             |
|                                                                                                                                    |  | Date of Review:30/05/2025               |
|                                                                                                                                    |  | Page <b>95</b> of <b>98</b>             |

|                                   |                                                                                                                |  |  |                                                                                                                                       |                                                               |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------|--|--|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
|                                   | 1.5gm)<br>Followed by<br>15mg/kg q12h                                                                          |  |  | redosing based on<br>trough levels                                                                                                    | levels<br>before 5th<br>dose to<br>achieve<br>15—<br>20mcg/ml |
| <b>Voriconazole<br/>IV and PO</b> | IV 6mg/kg<br>q12h x 2 doses,<br>then 4 mg/kg<br>q12hPO 400<br>mg q12 h x 2<br>doses then<br>200-300 mg<br>q12h |  |  | Normal but PO preferred, ideally<br>AVOID IV as solubilising agent may<br>lead to renal damage. If IV essential<br>monitor creatinine | AD                                                            |

. NB If anuric, morbidly obese or in acute renal failure, this equation will NOT give a true reflection of the creatinine clearance. If Anuric and oliguric (<500ml/day) patients assumed to have CrCl < 10ml/min. If morbidly obese, use ideal body weight (IBW): IBW (in kg) for males  
 = 50kg + 2.3kg for each inch over 5 feet (Devine formula –Average BMI of 23.0), IBW (in kg) for women =  
 49kg + 1.7kg for each inch over 5 feet (Robinson formula—average BMI of 21.1

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue: 30/06/2024               |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation: 20/05/2024            |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval: 30/04/2024            |
|                                                                                                                                    | Date of Review: 30/05/2025              |
| Page 96 of 98                                                                                                                      |                                         |

### 3.1 Annexure 1

#### PRE-AUTHORIZATION PRESCRIBING FORM - RESTRICTED ANTIMICROBIALS

This is a Pre-authorization form for Restricted Antimicrobial. This form must be completed when prescribing Restricted/Reserved Antimicrobials.

**Date:**

|                                                                          |                                                                    |                  |             |          |
|--------------------------------------------------------------------------|--------------------------------------------------------------------|------------------|-------------|----------|
| COLISTIN                                                                 | VANCOMYCIN                                                         |                  | POLYMYXIN B |          |
| LINEZOLID                                                                | DAPTOMYCIN                                                         |                  | CASPOFUNGIN |          |
| TIGECYCLINE                                                              | ANIDULAFUNGIN                                                      |                  | MINOCYCLIN  |          |
| TEICOPLANIN                                                              | FOSFOMYCIN                                                         |                  | AZTREONAM   |          |
| CEFTAZIDIME & AVIBACTAM                                                  | CEFTRIAZONE, DISODIUM EDETATE & SULBACTAM                          |                  |             |          |
| <b>Patient Information &amp; Diagnosis</b>                               |                                                                    |                  |             |          |
| Name:                                                                    |                                                                    | Age:             |             | Sex:     |
| Mr.No:                                                                   | Ward:                                                              | Weight:          | Sr.cr:      |          |
| Diagnosis:                                                               |                                                                    |                  |             |          |
| <b>Request for pre-authorized /Restricted Antimicrobial</b>              |                                                                    |                  |             |          |
| Date                                                                     | Antimicrobial (s) prescribed                                       | Dose & frequency | Route       | Duration |
|                                                                          |                                                                    |                  |             |          |
|                                                                          |                                                                    |                  |             |          |
|                                                                          |                                                                    |                  |             |          |
|                                                                          |                                                                    |                  |             |          |
| Indication:                                                              | Empiric in High Risk patients<br>Only sensitive drug in C&S report |                  |             |          |
| Description:                                                             |                                                                    |                  |             |          |
| Prescriber:                                                              |                                                                    |                  |             |          |
| Department:                                                              | Sign:                                                              |                  |             |          |
| <b>FOR USE BY CP/ICN</b>                                                 |                                                                    |                  |             |          |
| Organisms:                                                               | Suspected<br>Culture-proven                                        |                  |             |          |
| Concurrent Antimicrobials:                                               |                                                                    |                  |             |          |
| Date Stopped:                                                            |                                                                    |                  |             |          |
| Name & sign of the CP:                                                   |                                                                    |                  |             |          |
| <b>Please send completed form to Clinical Pharmacist/ICN/ Department</b> |                                                                    |                  |             |          |



|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue: 30/06/2024               |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation: 20/05/2024            |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval: 30/04/2024            |
|                                                                                                                                    | Date of Review: 30/05/2025              |
| Page 97 of 98                                                                                                                      |                                         |

### 3.2 Annexure 2: Rational-Antimicrobial Usage Monitoring Form

This is an antibiotic prescription monitoring tool. Be a part of the antimicrobial conservation movement.

|                                                                                            |                              |                  |                                     |           |
|--------------------------------------------------------------------------------------------|------------------------------|------------------|-------------------------------------|-----------|
| <b>Patient Information &amp; Diagnosis</b>                                                 |                              |                  | <b>Date:</b>                        |           |
| Name:                                                                                      |                              | Age:             |                                     | Sex:      |
| Mr.No:                                                                                     | Ward:                        | Weight:          | Sr.cr:                              |           |
| Diagnosis:                                                                                 |                              |                  |                                     |           |
| <b>Antimicrobial prescription details</b>                                                  |                              |                  |                                     |           |
| Date                                                                                       | Antimicrobial (s) prescribed | Dose & Frequency | Route                               | Duration  |
|                                                                                            |                              |                  |                                     |           |
|                                                                                            |                              |                  |                                     |           |
|                                                                                            |                              |                  |                                     |           |
|                                                                                            |                              |                  |                                     |           |
| <b>Rational usage assessment</b>                                                           |                              |                  |                                     |           |
| Empiric/Directed Antibiotic used:                                                          |                              |                  |                                     |           |
| Is the antibiotic according to institutional policy?                                       |                              |                  | <b>YES</b>                          | <b>NO</b> |
| Have the cultures been sent before the 1 <sup>st</sup> dose of antimicrobial?              |                              |                  | <b>YES</b>                          | <b>NO</b> |
| Has the antibiotics been reviewed based on culture report & de-escalated (after 48-72hrs)? |                              |                  | <input type="checkbox"/> <b>YES</b> | <b>NO</b> |
| Has the antibiotic been continued if the culture report is sterile? If so, why?            |                              |                  | <input type="checkbox"/> <b>YES</b> | <b>NO</b> |
| <b>If using three or more antimicrobials indicate reason for use:</b>                      |                              |                  |                                     |           |
| <b>If using antibiotics for &gt;10 day, indicate reasons:</b>                              |                              |                  |                                     |           |
| Was therapeutic duplication seen? Indicate reason:                                         |                              |                  | <b>YES</b>                          | <b>NO</b> |
| Was the drug being used in right dose and being adjusted for disease setting?              |                              |                  | <b>YES</b>                          | <b>NO</b> |
| Was the dosage adjusted for Hepatic/renal function ?                                       |                              |                  | <b>YES</b>                          | <b>NO</b> |
| Name & Sign: Clinical Pharmacist                                                           |                              |                  | Clinician/Prescriber                |           |

|                                                                                                                                    |                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
|  <b>KIMS ALSHIFA</b><br>SUPER SPECIALITY HOSPITAL | Doc No: KASPP/AB/01                     |
|                                                                                                                                    | Date of Issue:30/06/2024                |
| <b>Hospital Infection Control</b>                                                                                                  | Revision Date and Status: 30/05/2024/03 |
|                                                                                                                                    | Date of Creation:20/05/2024             |
| <b>ANTIBIOTIC POLICY</b>                                                                                                           | Date of Approval:30/04/2024             |
|                                                                                                                                    | Date of Review:30/05/2025               |
|                                                                                                                                    | Page <b>98</b> of <b>98</b>             |

| Initiated by                                                                        | Compiled by                                                                         |                                                                                      | Issued by                                                                             |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Swathi Laxmi A.P                                                                    | Dr. Vinitha Mary Joy                                                                | Dr. Mohammed Yahiya                                                                  | Dr. Abdulla Khaleel                                                                   |
| Quality Manager                                                                     | Clinical Microbiologist                                                             | Medical Superintendent                                                               | Clinical Coordinator                                                                  |
|  |  |  |  |